

InteractionGraphNet: A Novel and Efficient Deep Graph Representation Learning Framework for Accurate Protein–Ligand Interaction Predictions

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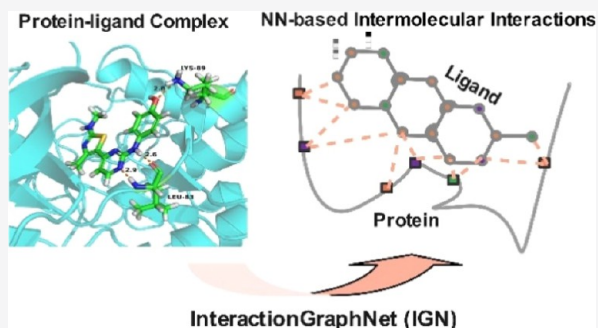
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ABSTRACT: Accurate quantification of protein–ligand interactions remains a key challenge to structure-based drug design. However, traditional machine learning (ML)-based methods based on handcrafted descriptors, one-dimensional protein sequences, and/or two-dimensional graph representations limit their capability to learn the generalized molecular interactions in 3D space. Here, we proposed a novel deep graph representation learning framework named InteractionGraphNet (IGN) to learn the protein–ligand interactions from the 3D structures of protein–ligand complexes. In IGN, two independent graph convolution modules are stacked to sequentially learn the intramolecular and intermolecular interactions, and the learned intermolecular interactions can be efficiently used for subsequent tasks.

Extensive binding affinity prediction, large-scale structure-based virtual screening, and pose prediction experiments demonstrated that IGN achieved better or competitive performance against other state-of-the-art ML-based baselines and docking programs. More importantly, such state-of-the-art performance was proven from the successful learning of the key features in protein–ligand interactions instead of just memorizing certain biased patterns from data.



INTRODUCTION

Accurate characterization of protein–ligand interactions has been one of the key challenges in structure-based drug design, whether for screening drug candidates from large compound libraries against targets of interest^{1–3} or for testing the potential undesired effects of compounds against possible off-target proteins.⁴ It has been well established that experimental determination of protein–ligand interactions is the most reliable way, but it is labor-extensive and costly in many cases. Owing to the revolution of computer-aided drug design (CADD) techniques and computational power, many theoretical methods, such as quantum mechanics/molecular mechanics (QM/MM) and free energy perturbation (FEP), are capable of accurately estimating protein–ligand interactions. Unfortunately, their routine applications in large-scale compound screening are hampered by extremely high computational overhead.^{5,6} On the contrary, molecular docking with the acceptable computational cost has been widely used in large-scale virtual screening (VS) of compounds against a variety of targets with known 3D structures, but its hit ratio is inherently nonideal as a result of the simplistic linear scoring functions (SF_s) implemented in molecular docking programs.⁷

To this end, numerous machine learning (ML)-based computational methods have been developed as alternatives

for protein–ligand interaction estimation.^{4,6,8–18} They can be roughly classified into two categories: traditional ML-based methods and deep learning (DL)-based methods. Similar to the quantitative structure–activity relationship (QSAR) in cheminformatics, a set of predefined features, such as interaction fingerprints and energy terms from classical SF_s, were extracted from protein–ligand complexes and used as the inputs of those traditional ML-based methods. Some representative methods, such as RFScore,^{19–21} NNscore,¹⁴ and deltaVinaRF,²² have shown higher flexibility and predictability in comparison with the classical SF_s. However, the predefined rules and/or descriptors might introduce bias of domain expertise and prevent an end-to-end learning manner from raw inputs.^{23–25} Therefore, much more attention of this field has been shifted to DL techniques because of their high expressive power and strong capacity to automatically extract

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task-specific features without any predefined rule and domain expertise.^{26,27}

Thus far, a wealth of DL methods has been developed for protein–ligand interaction predictions.^{4,6,8–13,15–18,28–36} While these methods show significant differences in network architectures and input representations, they can be classified into two major categories: structure-based and non-structure-based methods. In non-structure-based methods, no explicit 3D structural information such as binding structures or 3D atomic coordinates was used in model development. By contrast, they directly extracted the chemical information in two-dimensional (2D) space (SMILES strings/molecular graphs for small molecules and sequence strings/pocket graphs for proteins) by two detached modules, and the learned representation vectors were concatenated as the inputs for the subsequent task networks.^{4,18,30,31,33–35} Additionally, some coarse-grained 3D structural information such as protein distance map and spatial distance into bins were also attempted.^{18,33,35,37} Although some of these methods showed considerable performance and good computability, their generalization ability is still a big concern because the real molecular interactions occur in 3D space. As for the structure-based methods, the 3D convolutional neural network (CNN) has become a routine usage where a 3D rectangular grid encoded with various pieces of chemical information was used as the inputs of the 3D-CNN.^{10,11,13,15,16,28,29,32,38} It has been reported that many 3D-CNN models could yield better performance in binding affinity and pose predictions in comparison with traditional ML-based and classical SFs.^{10,11,13,15,16} However, they inherently suffer from several potential drawbacks. First, the superiority of the 3D-CNN trained on the 3D structural grid might boil down to the memorization of certain bias hidden in the dataset rather than successful generalization of protein–ligand interactions.^{39,40} Second, the use of 3D rectangular grid representation entails inefficient computations because the grid points are some kinds of high-dimensional and sparse 3D matrices.^{6,8,18} Therefore, there is a great demand for the development of computational models with affordable computability and high generalization ability.

Molecular graphs in which the connectivity between atoms in a molecule is represented as a graph are natural and effective representations for chemical molecules.^{25,41–43} To further improve the accuracy and generalization ability of DL-based methods on the prediction of protein–ligand interactions, a couple of algorithms built on molecular graphs embedded with 3D structural information have been proposed in the past few years.^{6,8,44} It is pleasurable to see that some key points for protein–ligand binding such as intermolecular recognition were explicitly characterized in these algorithms. For example, Lim et al. utilized a distance-aware graph attention algorithm to differentiate various types of intermolecular interactions.⁶ Feinberg et al. utilized a set of predefined rules on the Euclidean distance between atoms to distinguish edge types (covalent and noncovalent interactions).⁸ However, some potential shortcomings still exist. First, the predefined rules can lead to undesirable biases as mentioned earlier. Second, both Lim et al. and Feinberg et al. employed the ligand-based graph aggregation/gather manner to predict the interactions of complexes where only the states of ligand atoms were considered in the final predictions of complexes. Such operations may lead to overoptimistic statistics, especially for the biased DUD-E dataset. In addition, it should be noted that

the basic objective of these computational methods should not only give reliable predictions of the binding affinities and binding structures for experimentally determined complexes but also have good capability to enrich the true actives toward certain protein drug targets in large-scale VS. Unfortunately, our investigations revealed that only a few methods covered these two aspects.^{6,10,28}

Based on these observations, we proposed a novel and efficient deep graph learning framework named Interaction-GraphNet (IGN) to predict the interactions of protein–ligand complexes. A protein–ligand complex was represented as three independent molecular graphs embedded with 3D structural and chemical information, namely, ligand graph, protein graph, and bipartite protein–ligand graph. To be computationally efficient, only the protein atoms in the binding site were considered in graph construction, and the ligand graph and pocket graph shared the same graph convolution module (intramolecular graph convolution) to reduce the parameters employed in the IGN model because both were considered as the covalent connections in the corresponding graphs. Next, the representations of the ligand atoms and protein atoms from the ligand and protein graphs were automatically extracted through the node representation learning implemented by the intramolecular graph convolution module, and then the edge representation learning implemented by the intermolecular graph convolution module was sequentially stacked to extract the edge representations (learned the atom-pair interactions) in the bipartite protein–ligand graph where its initial atom states were directly inherited from the intramolecular graph convolution module. Then, the readouts from the edge representations in the bipartite protein–ligand graphs were directly used for making decisions accomplished by the downstream fully connected neural network (FCNN). The binding affinity prediction, large-scale [data volume larger than 1M (million)] structure-based VS (SBVS), and pose prediction tasks were tested for the IGN model. The results demonstrated that the IGN model can consistently outperform or match other state-of-the-art baselines and docking programs. Moreover, additional dedicated experiments and analyses also provided the evidence that the state-of-the-art predictions of the IGN model were contributed from successful learning of key features in protein–ligand interactions rather than the memorization of the hidden bias in the datasets. Moreover, further visualizations of the learned intermolecular interactions provide the interpretations that are well consistent with the expert knowledge.

It has been well recognized that graph convolution can considerably reduce the number of network parameters in comparison with typical FCNN. The checking of the employed parameters revealed that the IGN model is quite efficient and only needs 0.4M of parameters for the accurate binding affinity predictions for complexes, and the number of parameters is only approximately 1.4M even for large-scale (data volume larger than 1M) SBVS tasks. It is also a satisfactory trade-off between the complexity and performance of deep models compared with some commonly used alternatives in the chemistry community such as the MPNN.⁴⁵ Additionally, the size of the final well-trained IGN model for the binding affinity prediction and SBVS tasks is quite lightweight (1.5 and 5 MB for the binding affinity prediction and SBVS tasks, respectively). All in all, we believed that the IGN model can provide some kinds of workarounds for efficiently and

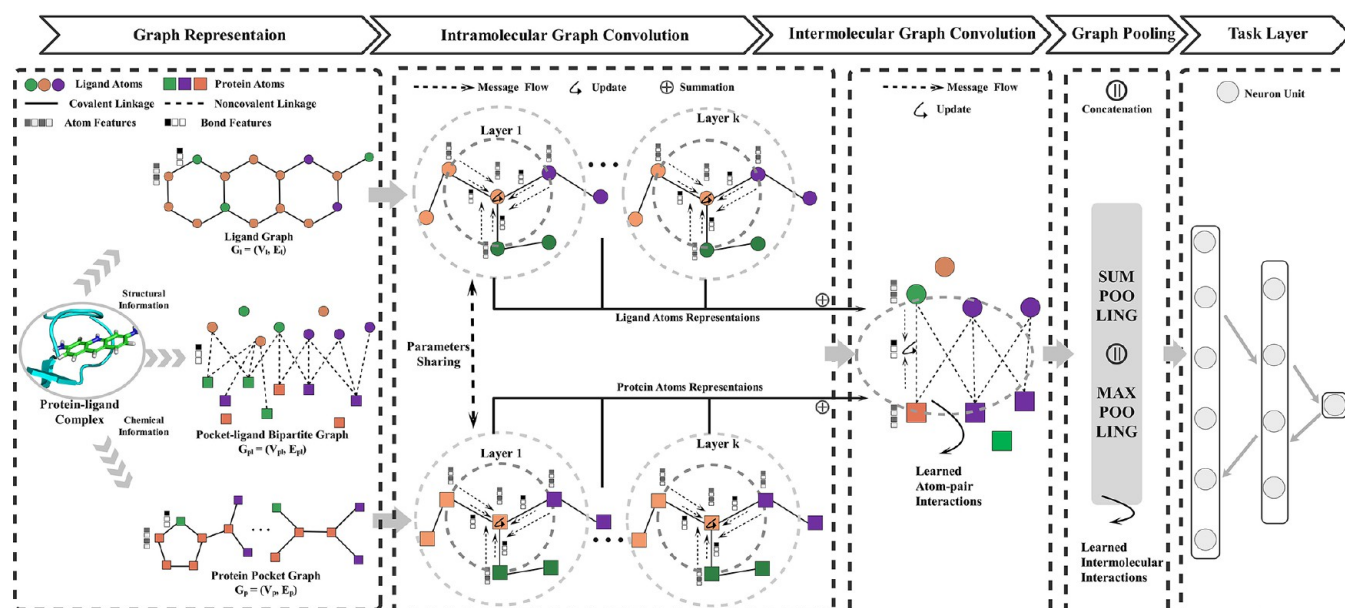


Figure 1. Detailed workflow of IGN, including graph representation, intramolecular graph convolution, intermolecular graph convolution, graph pooling, and task layer modules.

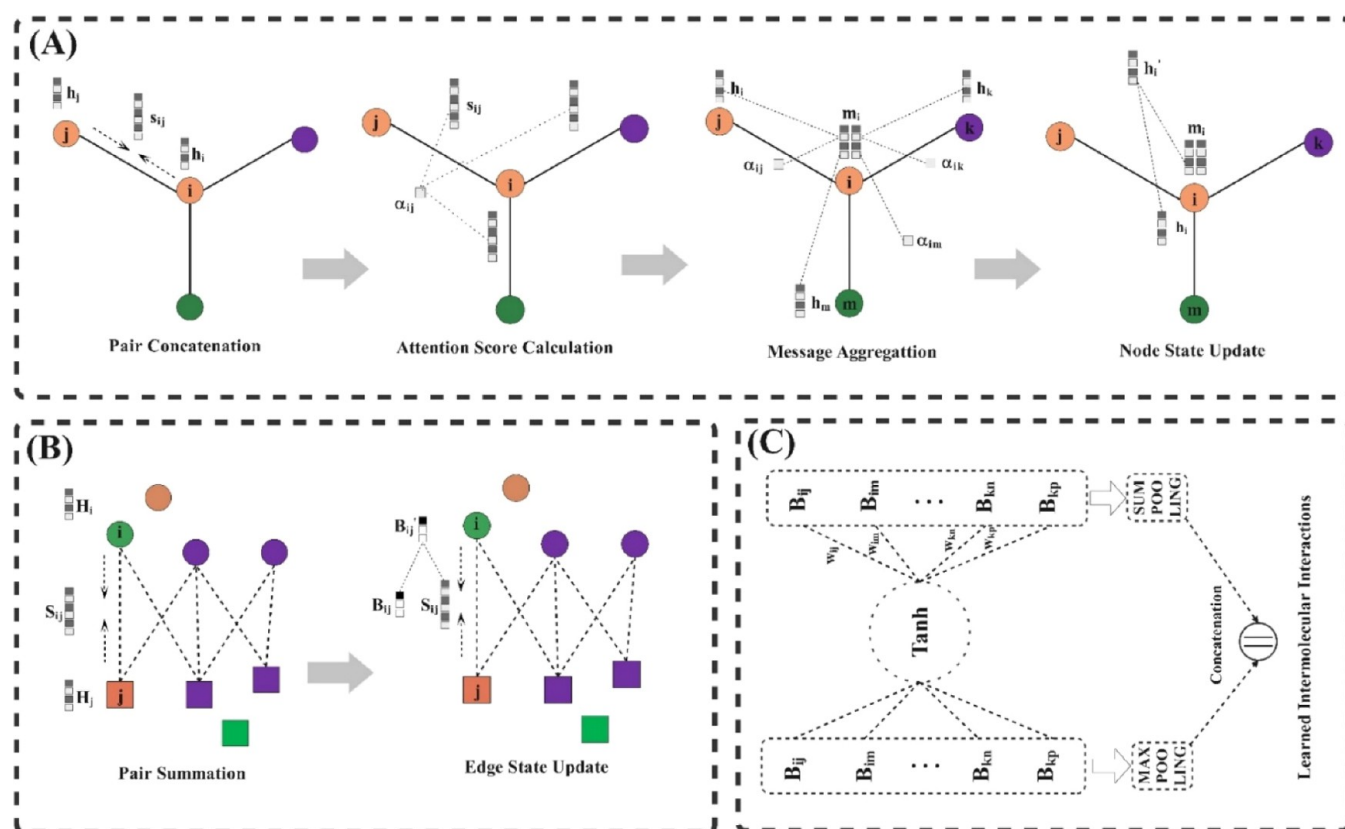


Figure 2. Picturesque visualizations of message passing in the intramolecular graph convolution module (A), intermolecular graph convolution module (B), and the details for graph pooling operation (C).

accurately predicting the binding of protein–ligand complexes with excellent generalization and interpretability.

RESULTS AND DISCUSSION

Performance Comparison with State-of-the-Arts on Binding Affinity Prediction. The binding affinity prediction assessment using the PDBBind dataset is a well-established

benchmark for evaluating the performances of protein–ligand interaction prediction models. Our IGN model was trained using the 8298 complexes from PDBBind V2016 and validated by the remaining 2067 complexes. As usual, two independent external test sets (the core sets from PDBBind V2016 and V2013) were employed to check the generalization ability of the IGN model (Figures 1 and 2). The statistics of the best

IGN model determined by the validation error minimum criterion are shown in Table 1. It can be observed that the IGN

Table 1. Performance Metrics for Different Datasets Given by the Best IGN Model Determined by the Validation Loss Minimum Criterion on Binding Affinity Prediction

PDBBind set	R_p	RMSE	MAE	R^2
training set ($N = 8298$)	0.987	0.309	0.221	0.974
validation set ($N = 2067$)	0.754	1.294	0.949	0.562
core set V2016 ($N = 262$)	0.837	1.220	0.940	0.687
core set V2013 ($N = 95$)	0.832	1.372	1.038	0.669

model achieves satisfactory metrics with $R_p > 0.750$ and root-mean-squared errors (RMSEs) of 1.294, 1.220, and 1.372 on the validation and two external test sets, respectively. To be more expressive, the experimental and predicted binding affinities for the training (Figure 3A), validation (Figure 3B), and two external test sets (Figure 3C,D) are presented in Figure 3. Apparently, the predicted and experimental values are highly correlated for the validation and two external test sets, highlighting the impressive predictive capacity of the IGN model. In order to explore the predictive capacity of the IGN model in depth, we analyzed several representative ML-based models on protein–ligand interaction predictions and listed

the corresponding statistics to the PDBBind V2016 and 2013 core sets in Table 2. For the sake of simplification, only the models that reported the results to either of these two core sets are listed, where the use of these two core sets as the external test sets is popular in community. As shown in Table 2, most of the reported models were developed based on the CNN algorithm, and some traditional ML methods such as RF and GBT were also highly favored, but less attention was paid to the GNN algorithm. For the core set of PDBBind V2016, our model ranks second ($R_p = 0.837$ and $RMSE = 1.220$) among the recent representatives listed in Table 2, and it is only worse than the extended connectivity interaction feature (ECIF)::GBT model ($R_p = 0.866$ and $RMSE = 1.169$) reported by Sánchez-Cruz et al.⁴⁶ It should be noted that the core set of PDBBind V2016 was taken as the validation set in the training of the ECIF::GBT model. However, in this study, this core set is totally unseen to the IGN model, and the best IGN model was determined by the independent validation error. In addition, it is generally accepted that all the ML models were equipped with the essence of data hunger. Therefore, it is not surprising that ECIF::GBT is superior to the IGN model with the assistance of the overoptimistic validation and a larger training dataset. Here, we reported excellent predictions ($R_p = 0.832$ and $RMSE = 1.372$) for the core set of PDBBind V2013. It should be noted that the size of

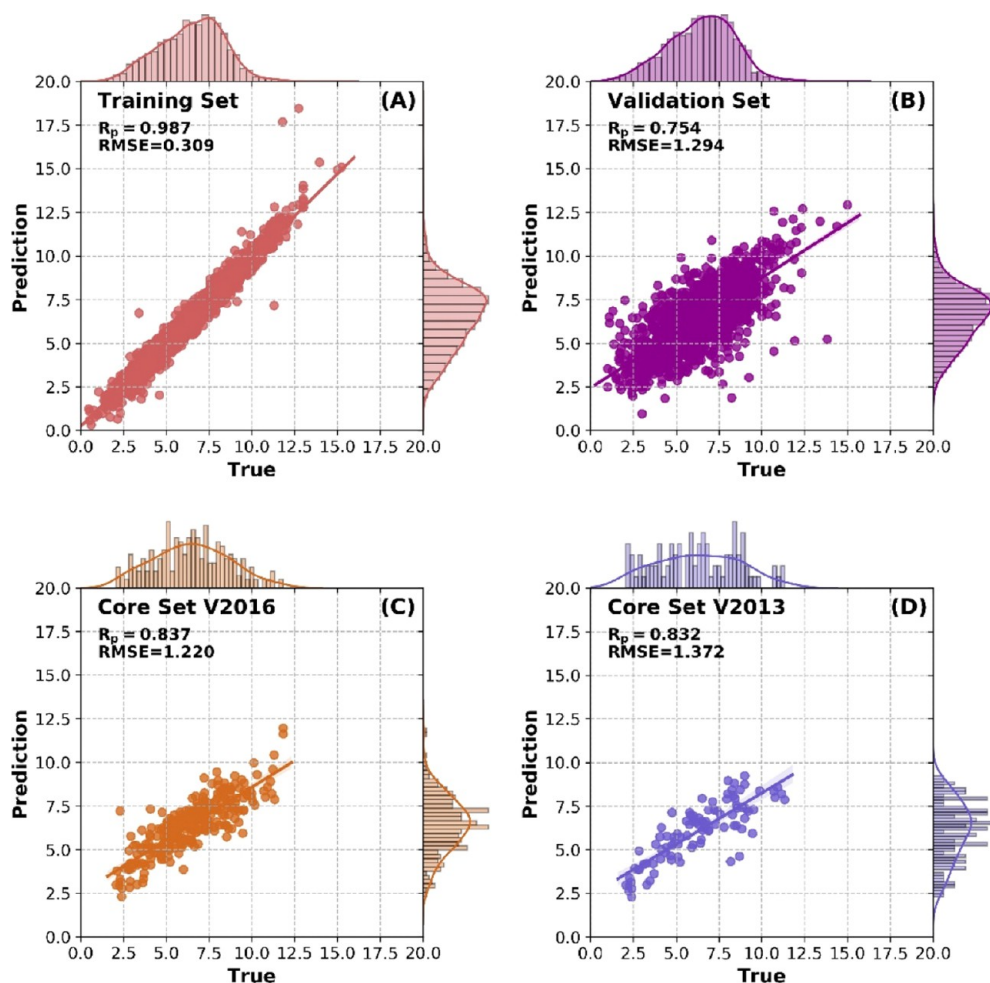


Figure 3. Correlation plot for training (A), validation (B), and two external test sets (C,D) in the PDBBind V2016 and V2013 datasets given by the best IGN model.

Table 2. Performance Comparison on the PDBBind V2016 and V2013 Core Sets of Recent Representative ML-Based Models for the Predictions of Protein–Ligand Interactions (the Top Three Models Were Bolded According to Their Statistics)^a

year	model	molecular representation	algorithm	V2016	V2013	training set
2021	IGN	molecular graph	GNN	R_p: 0.837 RMSE: 1.220 ($N = 262$)	R_p: 0.832 RMSE: 1.372 ($N = 95$)	PDBbind V2016 ($N = 8298$)
2020 ⁴⁶	ECIF::GBT	ECIF	GBT	R_p: 0.866 RMSE: 1.169 ($N = 285$)	NA	PDBbind V2016 and V2019 ($N = 9299$)
2020 ³⁸	FAST	molecular graph and 3D voxel representation	model fusion of 3D-CNN and GNN	R_p : 0.810 RMSE: 1.308 ($N = 290$)	NA	PDBbind V2016 ($N = 11\,717$)
2020 ¹⁷	NA	pairwise atomic properties	NN	R_p : 0.780 RMSE: 1.450 ($N = 134$)	NA	PDBbind V2018 ($N = 2675$)
2019 ⁹	OnionNet	distance-range-based atom-pair counts	2D-CNN	R_p : 0.816 RMSE: 1.278 ($N = 290$)	R_p: 0.782 RMSE: 1.503 ($N = 108$)	PDBbind V2016 ($N = 11\,906$)
2019 ⁴⁷	AGL-Score	algebraic graph descriptor	GBT	R_p: 0.833 RMSE: 1.271 ($N = 285$)	R_p: 0.792 RMSE: NA ($N = 195$)	PDBbind V2016 ($N = 3772$)
2019 ⁴⁸	PLEC-nn	protein–ligand extended connectivity	NN	R_p : 0.820 RMSE: NA ($N = 290$)	R_p : 0.770 RMSE: NA ($N = 195$)	PDBbind V2016 ($N = 12\,906$)
2018 ¹⁵	K _{Deep}	3D voxel representation	3D-CNN	R_p : 0.820 RMSE: 1.270 ($N = 290$)	NA	PDBbind V2016 ($N = 3767^{*24}$) ^b
2018 ¹¹	Pafnucy	3D voxel representation	3D-CNN	R_p : 0.780 RMSE: 1.420 ($N = 290$)	R_p : 0.700 RMSE: 1.620 ($N = 195$)	PDBbind V2016 ($N = 11\,906$)
2018 ⁴⁹	T-Bind	element-specific topological fingerprints	GBT	NA	R_p : 0.767 RMSE: 1.52 ($N = 195$)	PDBbind V2013 ($N = 2764$)
2018 ¹⁵	RFScore-V3	intermolecular contacts and Autodock Vina features	RF	R_p : 0.800 RMSE: 1.390 ($N = 290$)	NA	PDBbind V2016 ($N = 3767$)
2017 ⁵⁰	TNet-BP	multichannel topological fingerprints	1D-CNN	R_p : 0.810 RMSE: 1.340 ($N = 290$)	NA	PDBbind V2016 ($N = 3767$)

^aAll the models were presented by their optimal predictions on the PDBbind V2013 and V2016 core sets; abbreviations: GNN: graph neural network; NN: neural network; CNN: convolutional neural network; RF: random forest; GBT: gradient boosting tree; NA: not available; N represents the number of complexes. ^bThe datasets were augmented 24 times by rotation.

the core set of PDBbind V2013 tested here ($N = 95$) is far less than its original size ($N = 195$) as mentioned earlier. A similar situation was also reported by Zhu et al.¹⁷ and Zheng et al.⁹ Therefore, it is quite possible that the high prediction power of the IGN model to the core set of PDBbind V2013 was contributed from the metric calculation luckiness induced by the rapid reduction in the sample size. However, taking OnionNet as an example,⁹ the number of the tested complexes ($N = 108$) from the core set of PDBbind V2013 is comparable to that used in this study, but it only yields $R_p = 0.782$ and $RMSE = 1.503$, suggesting that such luckiness might not exist. Taken together, we believe that the IGN model can give reliable predictions for protein–ligand interactions.

To demonstrate the stability of the IGN model, the average metrics from three independent runs with different random seeds were also reported for the core set of PDBbind V2016. As can be seen from Supporting Information Table S1, the IGN model achieves an average R_p of 0.811 and a RMSE of 1.291 for the core set of PDBbind V2016, which are still competitive among the models listed in Table 2. As noted in the Experimental Section, a set of atomic coordinate-related geometric features were designed and encoded into the molecular graphs to better describe the spatial topology of a complex. To demonstrate the effectiveness of this set of descriptors, a simple comparison analysis was conducted by checking the model performance after removing these features. As shown in Supporting Information Table S2, it can be observed that the model performance clearly decreased after removing this set of descriptors compared with the original model (the best model or the average model), implying that the structural information in 3D space is quite essential for accurately predicting the binding of protein–ligand complexes. Interestingly, the IGN model without these complicated

geometric features (IGN_2D) seems more stable because its standard deviation (SD) of various performance indicators is obviously lower than that of the original IGN model, illustrating the outstanding stability of the IGN model predicted on the straightforward chemical information. From the perspectives of bias and variance theory, such a phenomenon is understandable because the enhancement of predictive performance (low bias) usually needs the expense of model stability (low variance), and it is irrational to seek a model with both low variance and low bias. From another perspective, a small perturbation on a complex structure may exert a direct impact on the geometric descriptors but not on the simple atom-/bond-level chemical descriptors.

Impacts of Structural and Sequence Similarity on the Performance of Binding Affinity Prediction. Recently, a couple of studies have verified the bias hidden in several well-established datasets, such as PDBbind and DUD-E, and therefore, it is not clear whether the claimed superiority of models was mainly contributed from the real signals of the data or from the hidden bias.^{39,40,51} Additionally, Li and Yang⁵² found that the binding affinity prediction capacity of ML-based models represented by RFScore¹⁹ was highly affected by the sample similarities between the training and external test sets, and they also observed that the ML-based models trained on dissimilar training samples did not outperform the classical SFs any more. Based on these reports, we examined the impacts of the protein and/or ligand similarities between the training and test sets on the IGN's performance.

For the protein similarity, both the sequence identity and structural similarity of the proteins were investigated. For each protein in the test set, NW-align⁵³ and TM-score⁵⁴ were used to identify the complexes in the training set with the maximal sequence identity and structural similarity, respectively. For the

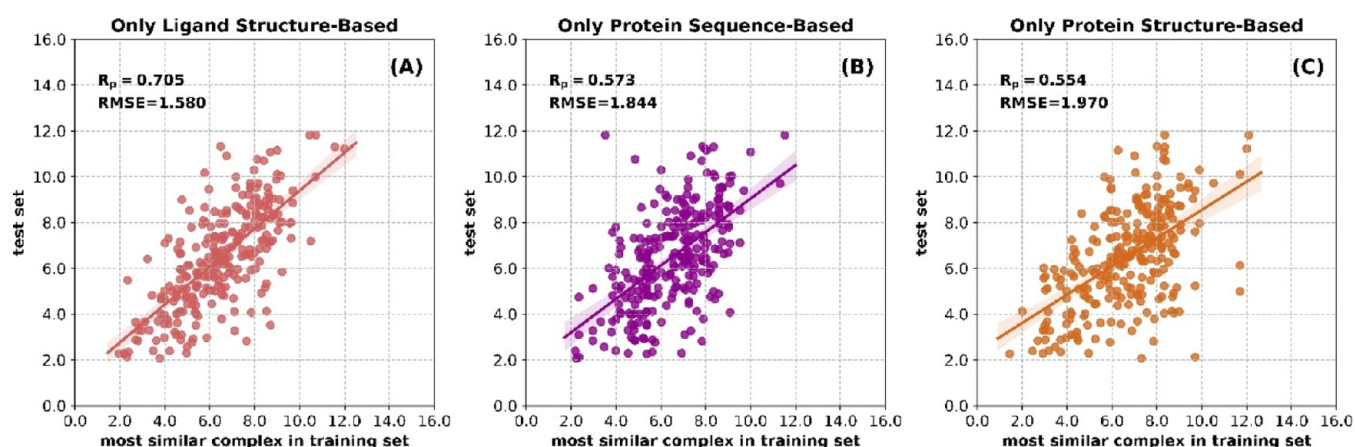


Figure 4. Maximum similarity criterion correlation plot given by ligand structure similarity (A), protein sequence identity (B), and protein structure similarity (C). The X-axis is the binding affinity data of most similar complexes in the training set, and the Y-axis is the corresponding binding affinity data for the test set.

ligand similarity, the Morgan fingerprints with 1024 bits at a radius of 2 were used to identify the complex in the training set with the maximum ligand structural similarity. Such analyses discovered that, when considering the sequence identity and structural similarity of proteins, most complexes (over 85%) in the test set have almost exactly matched partners in the training set (similarity score ≥ 0.990) but that for the ligand similarity is 24%. Such findings are not surprising because multiple coligand structures for the same protein with different assigned entries are available in the PDBBind database, but the structures of the ligands vary from one to another.

Here, the correlations between the complexes in the test set and their corresponding complexes with the highest similarity scores in the training set are presented in Figure 4. For the protein sequence identity and structural similarity (Figure 4B,C), the correlation coefficients can reach a moderate level with $R_p = 0.550$. In other words, one can give decent predictions on the binding affinity of a complex in the test set only using the binding affinity data from its most similar complex in the training set without any learning process. Strikingly, the correlation coefficient for the ligand similarity can reach a satisfactory level of $R_p = 0.705$ (Figure 4A), which even outperforms most classical SF_s . These observations illustrate that an intrinsic domain bias may exist in the PDBBind dataset and it is necessary to establish the corresponding baselines when developing ML-based models. However, fortunately, the predictive performance of the IGN model (Table 1) is far better than those baselines shown in Figure 4.

To further evaluate the impacts of the protein and/or ligand similarity on the IGN's performance, we removed similar complexes from the test set according to different levels of ligand similarity (protein sequence identity and structural similarity were not considered because most scores were close to 1.0 as mentioned before). The new metrics for the remaining dissimilar complexes in the test set were calculated and compared with the original metrics. In order to eliminate the impact of the size reduction of the test set, we did random sampling 100 times to generate 100 equal-sized test sets at different levels of the ligand similarity as the control and the average metrics are listed in Table 3. It can be observed that the performances for the control are stable with $R_p \approx 0.835$ and $RMSE \approx 1.220$, regardless of the test set size. However,

Table 3. Impacts of Ligand Similarity on the IGN's Performance^a

cutoff	no. of test set	R_p		RMSE	
		IGN	random	IGN	random
1.000	262	0.837	0.837	1.220	1.220
0.990	199	0.816	0.835	1.304	1.223
0.970	197	0.815	0.835	1.309	1.223
0.950	197	0.815	0.835	1.309	1.223
0.930	195	0.829	0.834	1.267	1.223
0.910	194	0.836	0.834	1.248	1.223
0.890	193	0.832	0.834	1.251	1.222
0.870	193	0.832	0.834	1.251	1.222
0.850	192	0.833	0.834	1.253	1.223
0.830	185	0.836	0.836	1.249	1.221
0.810	178	0.835	0.836	1.258	1.219
0.790	173	0.835	0.835	1.261	1.221
0.770	169	0.832	0.836	1.261	1.219
0.750	157	0.831	0.836	1.264	1.216
0.730	151	0.843	0.836	1.246	1.216
0.710	146	0.843	0.836	1.244	1.216
0.690	136	0.839	0.835	1.277	1.219
0.670	125	0.835	0.835	1.298	1.218
0.650	121	0.834	0.835	1.313	1.219
0.630	115	0.823	0.835	1.327	1.219
0.610	104	0.819	0.835	1.331	1.216

^aTaking line two as an example, 0.990 represents the ligand similarity level where the complexes in the test set with the highest similarity score ≥ 0.990 were removed and the number of the remaining dissimilar complexes in the test set is 199.

the performance of the IGN model on average becomes slightly worse when the similar complexes in the test set are removed gradually, especially for the RMSE metric, indicating that the IGN model gives relatively more unreliable predictions for the dissimilar complexes in the test set in comparison with the similar complexes. The work reported by Li et al. concluded that the similarity can heavily affect the performance of ML-based models such as RFscore. Different from their conclusion, our analyses demonstrate that such impact is not very significant for the IGN model. Here, it can be observed that the R_p values on some levels of ligand similarity (especially cutoff = 0.730 and 0.710) are even higher than those of the whole test set. Such observation may be caused by metric

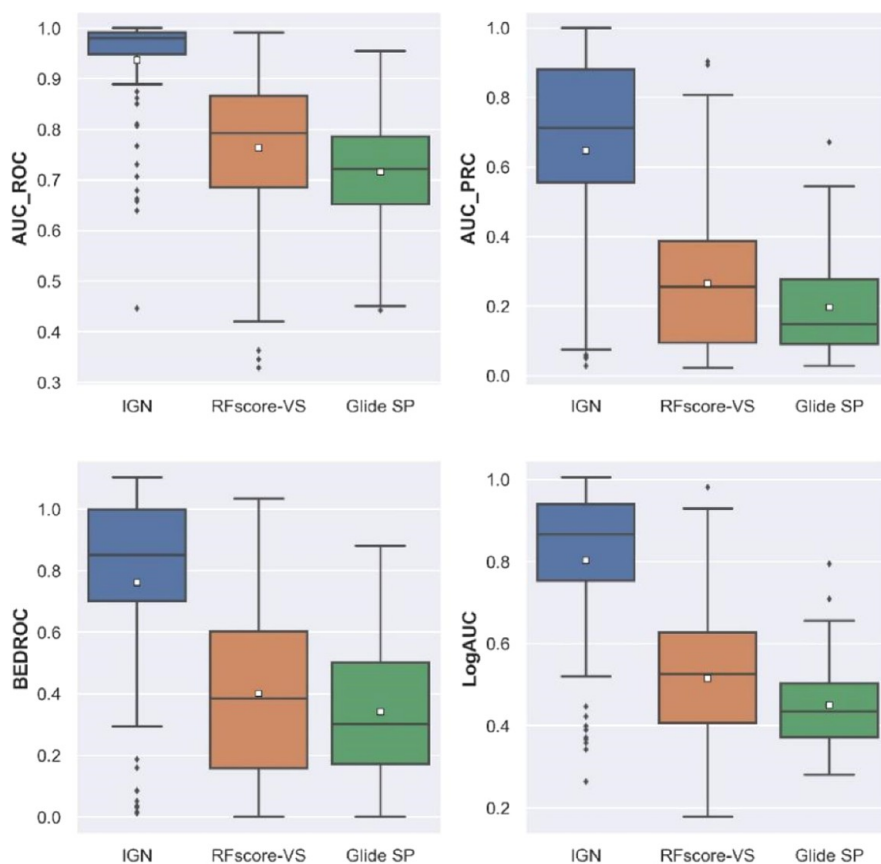


Figure 5. Performance indicators including AUC_ROC, AUC_PRC, BEDROC, and LogAUC distribution plot for each protein in the DEKOIS2.0 dataset given by IGN, RFscore-VS, and Glide SP.

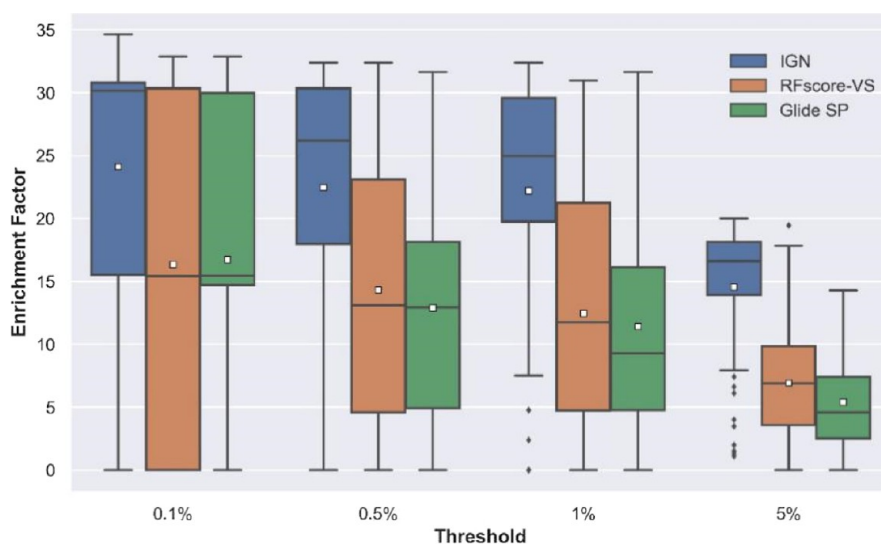


Figure 6. EF distribution plotted at different thresholds including 0.1, 0.5, 1, and 5% for each protein in the DEKOIS2.0 dataset given by IGN, RFscore-VS, and Glide SP.

calculation luckiness where relatively bad RMSE values do not always mean relatively bad correlations (R_p) in comparison with the original reports.

Motivation of the Assessment of Screening Power for IGN. The abilities of sophisticated ML-based models to enrich true actives in a large-scale compound library screening (VS power) should be tested as described in many studies.^{51,55,56} Unfortunately, our investigations revealed that

such testing was rarely explored by most recently proposed ML-based models.^{9,11,13,15,17,19,38,44,48–50,57,58} In addition, the recent systematic assessment of 14 ML-based models illustrated that these sophisticated models could hardly achieve satisfactory results in large-scale VS, and they could even not outperform classical SFs such as Glide SP.⁵⁶ Based on these observations, the VS power of the IGN model was systematically evaluated using the DUD-E, DEKOIS2.0, and LIT-PCBA

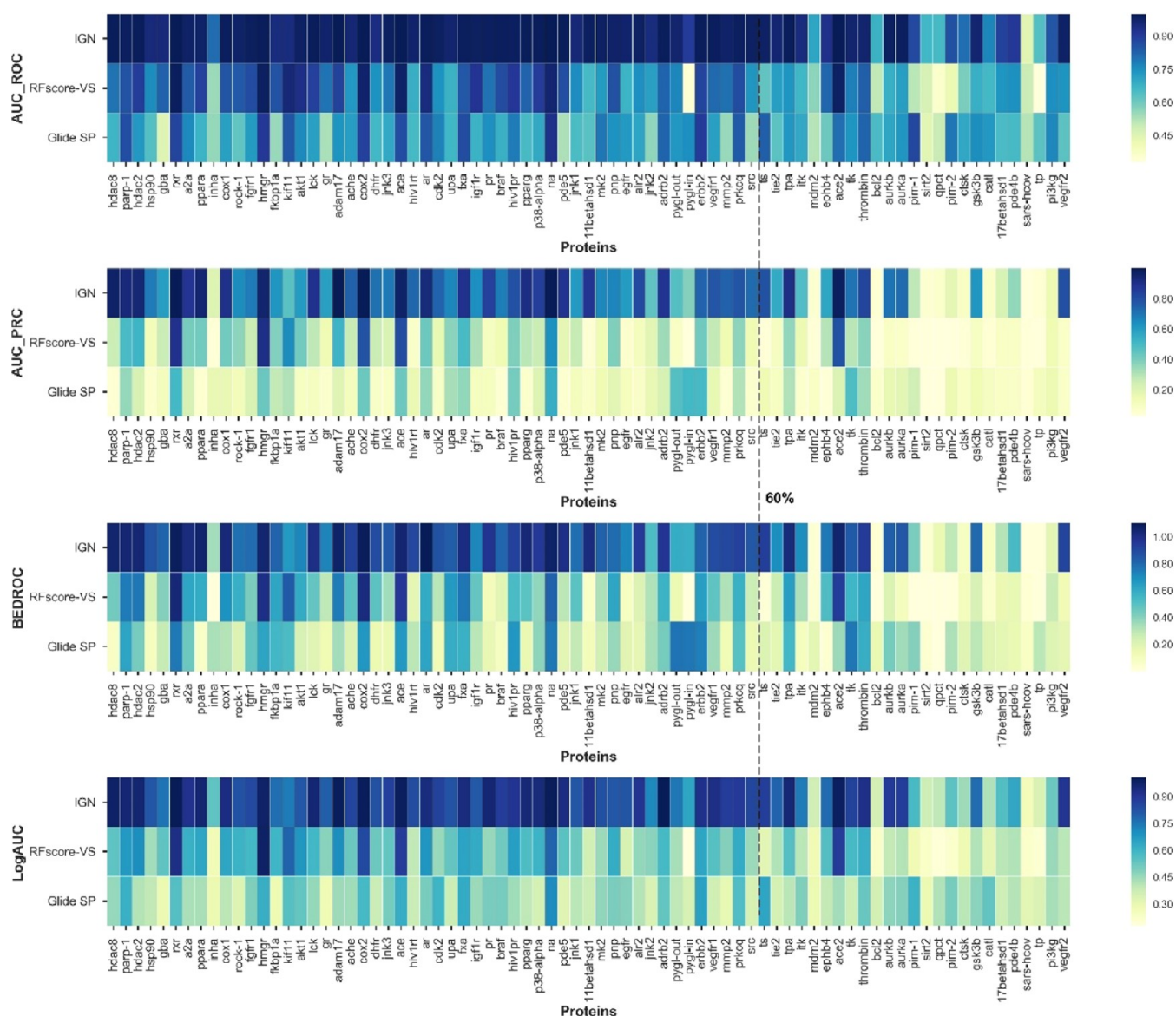


Figure 7. AUC_ROC, AUC_PRC, BEDROC, and LogAUC heat map plot for each protein in the DEKOIS2.0 datasets given by IGN, RFscore-VS, and Glide SP. The proteins are sorted by their sequence identity with proteins in DUD-E in the descending order. The proteins in the left of the dotted line have a sequence identity over 60% while those on the right have their scores below 60%.

datasets. In the first evaluation stage, a target-free model was trained and validated by the DUD-E dataset and then tested by the DEKOIS2.0 dataset. Different from the practices in many previous studies,^{6,18} we did not use the DUD-E dataset as the external test set due to the potential hidden bias in this dataset. In the second evaluation stage, three more challenging and representative targets from the LIT-PCBA dataset with the ratio of decoys to actives higher than 1000 were selected to better mimic the real VS scenarios. The models trained on this stage were target-specific and they were externally tested by the corresponding test sets provided by the original paper.⁵⁹ In the third evaluation stage, the IGN model trained on the DUD-E dataset was also tested on three challenging targets in the LIT-PCBA dataset for a deeper verification of the generalization ability of IGN. RFscore-VS and Glide SP were used as the baselines due to their superiority in SBVS according to the recent study reported by Shen et al.⁵⁶ Overall, RFscore-VS was the top-performing ML-based model among the 14 tested algorithms and it could also match the Glide SP docking

method in large-scale VS. The RFscore-VS features for the datasets were calculated by the Open Drug Discovery Toolkit (ODDT) Python package,⁶⁰ and model training was implemented by the scikit-learn Python package. In the training of the corresponding RFscore-VS models, each one was customized to optimize around the hyperparameter space provided by the original paper using the corresponding validation set for each experiment, and in a similar way, we reported the best results according to the validation set error minimum criterion. The original DEKOIS2.0 dataset contains 81 diverse targets, where four targets (i.e., cyp2a6, er-beta, jak3, and pdk1) could not be processed by the RFscore-VS feature generation algorithm in the ODDT Python package. Therefore, the remaining 77 targets were used for the follow-up analysis.

Assessment of Screening Power on DEKOIS2.0. Here, the box plots of various performance indicators given by IGN, RFscore-VS, and Glide SP are shown in Figure 5 (AUC_ROC, AUC_PRC, BEDROC, and LogAUC) and Figure 6 (EFs at

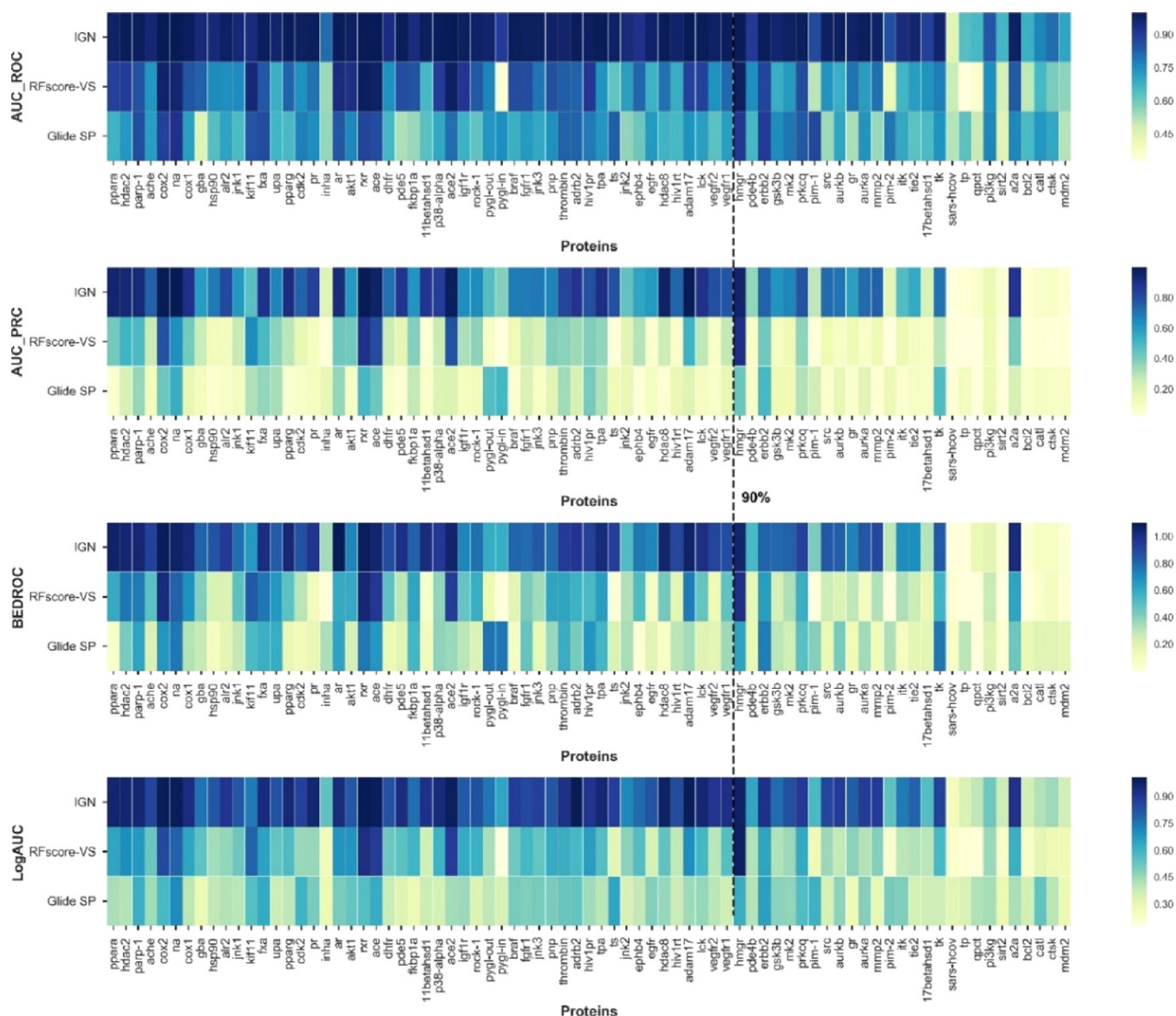


Figure 8. AUC_ROC, AUC_PRC, BEDROC, and LogAUC heat map plot for each protein in the DEKOIS2.0 dataset given by IGN, RFscore-VS, and Glide SP. The proteins are sorted by their structural similarities with proteins in DUD-E in the descending order. The proteins in the left of the dotted line have a structure similarity over 90% while those on the right have their scores below 90%.

different thresholds including 0.1, 0.5, 1, and 5%). The detailed statistics of various performance indicators for each target in DEKOIS2.0 given by the three models are available in [Supporting Information Tables X1–X3](#). As can be seen in [Figures 5 and 6](#), RFscore-VS performs slightly better than Glide SP on various performance indicators, which is consistent with our recent study.⁵⁶ However, the IGN model obviously outperforms RFscore-VS and Glide SP on this benchmark dataset. Specifically, the IGN model gives satisfactory average predictions for the 77 targets in DEKOIS2.0 with an AUC_ROC of 0.938, an AUC_PRC of 0.646, a BEDROC of 0.762, a LogAUC of 0.803, an $EF_{0.1\%}$ of 24.18, an $EF_{0.5\%}$ of 22.46, an $EF_{1\%}$ of 22.23, and an $EF_{5\%}$ of 14.58. However, the corresponding average predictions given by RFscore-VS are 0.764, 0.266, 0.401, 0.516, 16.34, 14.31, 12.46, and 6.94 and those given by Glide SP are 0.709, 0.189, 0.332, 0.445, 16.81, 12.67, 11.17, and 5.25, respectively, highlighting the great predictive capacity of the IGN model in VS. Here, compared with the indicators such as the area under

curve (AUC_ROC, AUC_PRC, BEDROC, and LogAUC), the EFs measured by counting the total number of the interesting compounds among the top-ranked compounds are more intuitive and meaningful. Gratifyingly, the average EFs at the thresholds of 0.1, 0.5, 1, and 5% provided by IGN were correspondingly improved approximately 30–50% points in comparison with those provided by RFscore-VS and Glide SP. Additionally, according to the performance indicators on DEKOIS2.0 from three independent runs given by the IGN model in [Supporting Information Table X3](#), the above conclusion can be made.

To be more persuasive, the Friedman test was employed to check the hypothesis whether the performances of the three compared models are at the same level (significance level = 0.05). As shown in [Supporting Information Table S3](#), the p -values of different metrics on DEKOIS2.0 for the three compared models are all lower than 0.05, suggesting that the performances of the three models are not at the same level. Subsequently, the posthoc Nemenyi test (significance level =

Table 4. Mean Performance Indicators Including AUC_ROC, AUC_PRC, BEDROC, LogAUC, EF_{0.1%}, EF_{0.5%}, EF_{1%}, and EF_{5%} for Different Protein Families in the DEKOIS2.0 Dataset Given by IGN, RFscore-VS, and Glide SP

Family	Model	AUC_ROC	AUC_PRC	BEDROC	LogAUC	EF _{0.1%}	EF _{0.5%}	EF _{1%}	EF _{5%}
GPCR	IGN	0.989	0.882	0.991	0.961	29.74	27.34	27.34	18.09
	RFscore-VS	0.831	0.373	0.598	0.598	29.74	25.15	19.02	9.17
	Glide SP	0.793	0.225	0.443	0.482	7.68	16.17	13.10	7.19
Hydrolase	IGN	0.940	0.676	0.808	0.809	26.74	25.74	25.32	14.07
	RFscore-VS	0.792	0.281	0.422	0.538	19.49	15.17	13.32	7.28
	Glide SP	0.662	0.161	0.280	0.426	16.92	10.85	9.07	4.28
Isomerase	IGN	0.949	0.620	0.818	0.767	32.39	32.39	32.39	11.49
	RFscore-VS	0.855	0.387	0.618	0.596	32.39	32.39	24.92	7.84
	Glide SP	0.554	0.277	0.549	0.287	32.39	27.77	22.43	5.75
Kinase	IGN	0.965	0.627	0.755	0.817	24.56	21.93	21.76	15.51
	RFscore-VS	0.766	0.204	0.344	0.494	15.81	12.70	10.37	6.49
	Glide SP	0.716	0.162	0.296	0.458	16.39	11.85	10.63	4.67
Ligase	IGN	0.707	0.057	0.036	0.367	0.00	0.00	0.00	2.00
	RFscore-VS	0.565	0.041	0.037	0.278	0.00	0.00	0.00	2.00
	Glide SP	0.518	0.093	0.180	0.291	29.95	9.98	4.99	2.00
Nuclear Receptor	IGN	0.988	0.870	0.997	0.937	27.82	26.36	25.41	17.89
	RFscore-VS	0.875	0.378	0.525	0.626	14.23	15.74	13.62	8.92
	Glide SP	0.724	0.195	0.319	0.446	9.32	8.61	8.58	4.78
Other	IGN	0.679	0.053	0.032	0.358	0.00	0.00	0.00	1.48
	RFscore-VS	0.505	0.032	0.002	0.232	0.00	0.00	0.00	0.49
	Glide SP	0.652	0.074	0.160	0.357	0.00	4.37	4.70	3.95
Oxidoreductase	IGN	0.950	0.710	0.837	0.835	25.60	24.51	23.54	15.38
	RFscore-VS	0.783	0.335	0.453	0.562	12.37	13.57	12.98	7.95
	Glide SP	0.717	0.194	0.321	0.428	19.89	11.89	9.06	5.58
Protease	IGN	0.909	0.666	0.749	0.796	23.25	21.70	22.63	14.33
	RFscore-VS	0.784	0.366	0.532	0.566	20.70	19.39	18.33	8.46
	Glide SP	0.735	0.240	0.413	0.455	20.65	14.88	14.35	6.70
Transferase	IGN	0.873	0.482	0.590	0.684	19.67	18.73	17.81	11.48
	RFscore-VS	0.591	0.156	0.222	0.376	10.91	8.71	7.71	3.63
	Glide SP	0.722	0.259	0.437	0.466	15.52	17.83	15.44	6.65

0.05) was further used to check whether the difference between any two of the compared models is statistically significant. As shown in Supporting Information Table S4, the differences between IGN and the other two models (RFscore-VS and Glide SP) are statistically significant for all the considered metrics but that between RFscore-VS and Glide SP is not significant, further implying that the IGN model has its great superiority in large-scale VS compared with the state-of-the-art methods including RFscore-VS and Glide SP.

We also investigated the impact of the sequence identity and structural similarity of proteins on the VS performances for different models. The sequence identity and structural similarity were also computed by NW-align and TM-Score. The corresponding results given by four indicators including AUC_ROC, AUC_PRC, BEDROC, and LogAUC are presented in Figure 7 (sequence identity) and Figure 8 (structural similarity). As for the sequence identity, it seems that the ML-based models (IGN and RFscore-VS) can give more reliable predictions for the proteins similar to the training proteins in comparison with those dissimilar to the training

proteins, and this phenomenon is not so obvious for the classical SF_s represented by Glide SP. Taking the sequence identity of 60% in Figure 7 as an example, the average AUC_ROC values for 52 similar and 25 dissimilar proteins given by IGN, RFscore-VS, and Glide SP are 0.977 and 0.857, 0.820 and 0.646, and 0.720 and 0.685, respectively. A similar observation can be also concluded from the structural similarity shown in Figure 8 (the corresponding threshold is 90% as most structural similarity scores are close to 1.0). These observations suggest that the sophisticated ML-based models can perform well for the proteins that have been frequently identified before but may perform comparatively unsatisfactorily to less popular or novel proteins in comparison with similar proteins. Apparently, it is not surprising for such scenarios because the interpolation capacity of ML-based models is comparatively better than their extrapolation capability in many cases. However, the classical SF_s driven by force field or physics are not easily trapped into such sceneries. Hopefully, such limitations can be mitigated with the

Table 5. Performance Comparison of IGN (Target-Specific), IGN (DUD-E), RFscore-VS (Target-Specific), and Glide SP on the Three Representative Targets in the Challenging LIT-PCBA Dataset

model	AUC_ROC	AUC_PRC	BEDROC	LogAUC	EF _{0.1%}	EF _{0.5%}	EF _{1%}	EF _{5%}
FEN1								
IGN (target-specific)	0.882	0.033	0.295	0.635	119.255	41.290	30.424	10.651
IGN (DUD-E)	0.668	0.004	0.088	0.416	21.683	13.039	8.693	4.130
RFscore-VS (target-specific)	0.713	0.018	0.154	0.372	65.048	26.078	14.126	5.217
Glide SP	0.584	0.001	0.038	0.316	0.000	2.173	2.173	4.130
model	AUC_ROC	AUC_PRC	BEDROC	LogAUC	EF _{0.1%}	EF _{0.5%}	EF _{1%}	EF _{5%}
KAT2A								
IGN (target-specific)	0.504	0.001	0.030	0.299	0.000	4.157	4.162	1.666
IGN (DUD-E)	0.610	0.002	0.076	0.378	41.196	12.472	6.243	2.916
RFscore-VS (target-specific)	0.511	0.001	0.013	0.159	0.000	0.000	2.081	1.250
Glide SP	0.493	0.001	0.003	0.298	0.000	0.000	0.000	0.833
model	AUC_ROC	AUC_PRC	BEDROC	LogAUC	EF _{0.1%}	EF _{0.5%}	EF _{1%}	EF _{5%}
PKM2								
IGN (target-specific)	0.701	0.005	0.044	0.401	0.000	3.322	2.496	3.499
IGN (DUD-E)	0.528	0.003	0.028	0.311	0.000	1.661	0.832	1.833
RFscore-VS (target-specific)	0.610	0.003	0.026	0.247	8.304	1.661	1.664	1.833
Glide SP	0.519	0.002	0.009	0.302	0.000	0.000	0.000	0.971

accumulation of more experimental data or by integrating some physical laws in the future.

The proteins in the DEKOIS2.0 dataset can be classified into 10 protein families. The mean performance indicators including AUC_ROC, AUC_PRC, BEDROC, LogAUC, EF_{0.1%}, EF_{0.5%}, EF_{1%}, and EF_{5%} for each protein family given by IGN, RFscore-VS, and Glide SP are presented in Table 4. It can be observed that the IGN model can achieve the best predictions for most families. However, for the ligase and other families, both the IGN and RFscore-VS models perform inferiorly compared with Glide SP. Such poor performance might be caused by the introduction of dissimilar proteins into DEKOIS2.0. More precisely, the ligase family only contains the mdm2 protein, and the other family only contains the bcl2 protein. The highest scores of the sequence identity and structural similarity for mdm2 are very low for all the proteins in DUD-E, as shown in Figures 7 and 8 (0.424 for the sequence identity and 0.340 for the structural similarity) and those for bcl2 are also very low (0.387 for the sequence identity and 0.402 for the structural similarity). Therefore, it is acceptable that the predictions of IGN and RFscore-VS for these two families are worse than those of Glide SP.

Assessment of Screening Power on the Unbiased and More Challenging LIT-PCBA Dataset. For many years, the DUD-E dataset has been regarded as a gold standard for benchmarking the VS power of ML-based methods, but many recent investigations unambiguously demonstrated that the hidden bias of the artificially generated decoys would lead to an overestimation on the prediction accuracy of various ML-based methods in VS.^{39,40} More recently, a novel dataset called LIT-PCBA was designed for benchmarking VS and ML,⁵⁹ and all the actives and decoys in LIT-PCBA were taken from the experimental data (PubChem BioAssay database) under homogeneous conditions and the high imbalance between actives and decoys was considered for mimicking the “real-life” screening decks better. The preliminary testing for various state-of-the-art methods including 2D similarity, 3D shape matching, and docking indicated that it is an extremely challenging dataset.⁵⁹ More importantly, the provided ligand set was unbiased by the recent asymmetric validation embedding (AVE) procedure. To further demonstrate the

reliability and robustness of IGN in VS, three representative targets in LIT-PCBA (i.e., FEN1, KAT2A, and PKM2) were selected to build the target-specific IGN models as noted before.

As shown in Table 5, it can be recognized that the performances on these three targets are apparently unimpressive compared with those on the DEKOIS2.0 dataset. The best AUC_ROC and BEDROC values for the three targets are 0.882 and 0.295, respectively, which are much lower than those for the DEKOIS2.0 dataset, highlighting the great difficulty to effectively distinguish the actives and inactives in LIT-PCBA. Here, the well-known docking program, Glide SP, only shows slightly better predictive capacity than random predictions, particularly for KAT2A and PKM2. As for the more intuitive EFs, Glide SP almost fails to give any enrichment for the actives in the 0.1, 0.5, and 1% top-ranked compounds for these two targets. Actually, for the target KAT2A, all the three methods almost yield random predictions in terms of the AUC_ROC metric, but fortunately, the IGN model can make more decent predictions in terms of EFs compared with the other two methods. Overall, RFscore-VS outperforms the well-known Glide SP docking program, especially on the FEN1 target, and it also outperforms the IGN model on certain indicators for certain targets, such as the EF_{0.1%} metric for the PKM2 target, but taken together, the IGN model can still keep its competitive predictive capacity among the baselines for all the three targets. Concretely, the IGN model doubles EFs compared with RFscore-VS for the FEN1 target and achieves better predictions for the other two targets according to the AUC metrics. Additionally, an investigation of the original paper⁵⁹ illustrated that the reported AUC_ROC values for these three targets given by the one-nearest-neighbor similarity search method are all close to 0.5, demonstrating the competitive predictive ability of the IGN model for this challenging dataset. In a similar way, the detailed performance indicators for the three targets from three independent runs given by the IGN model are provided in Supporting Information Table X4, and a similar conclusion that the IGN model performs the best on average among the compared baselines can also be reached based on these new indicators.

Table 6. Performance Comparison on the Test Set (Core Set of PDBBind V2016) Given by Different IGN Models Trained on the Original Complex (IGN), Ligand-Only (IGN_Ligand), and Protein-Only (IGN_Protein) Datasets of PDBBind

type	model	R_p	RMSE	MAE	R^2
best model	IGN	0.837	1.220	0.940	0.687
	IGN_Ligand	0.781	1.369	1.062	0.606
	IGN_Protein	0.791	1.339	1.027	0.623
average model	IGN	0.811 ± 0.028	1.291 ± 0.073	0.990 ± 0.047	0.649 ± 0.040
	IGN_Ligand	0.773 ± 0.010	1.390 ± 0.025	1.075 ± 0.013	0.593 ± 0.015
	IGN_Protein	0.771 ± 0.019	1.395 ± 0.050	1.068 ± 0.044	0.591 ± 0.029

Here, different VS datasets may possess different data patterns, and training the target-specific IGN models for different targets in the LIT-PCBA dataset is also not so practical in real-life applications. To further verify the generalization ability of IGN, we directly tested the performance of the IGN models trained on DUD-E for the three challenging targets in the LIT-PCBA dataset [Table 5, IGN (DUD-E) model]. As can be seen in Table 5, the IGN (DUD-E) models became worse on two targets (FEN1 and PKM2) but much better on KAT2A compared with its target-specific counterpart. Unlike the Glide SP module, the IGN (DUD-E) model can still give satisfactory enrichments for all the three targets, and its performance is also comparable to the target-specific RFscore-VS model, indicating that the IGN model is equipped with decent generalization ability.

High Performance of IGN Was Not Contributed from Data Bias. Many recent investigations have demonstrated that the DL models such as the 3D-CNN trained on the 3D binding structures of protein–ligand complexes achieve similar performances compared with those trained on the datasets only containing ligand structures or protein structures,^{39,40} highlighting that these DL models may make decisions by merely memorizing certain hidden ligand patterns or protein patterns instead of learning and generalizing protein–ligand interaction patterns. On this ground, we further explored whether the state-of-the-art performance of the IGN model was affected by such issues or not. More concretely, we first trained the ligand-only and protein-only IGN models based on the PDBBind dataset and compared them with the original complex-based model. For the ligand-only model, all the protein information in the protein graphs $G_p = (V_p, E_p)$ and the distance information in the original pocket–ligand graphs $G_{pl} = (V_{pl}, E_{pl})$ were masked by zeros before sending to the task-layer module to approximate that the IGN model only utilizes the ligand information to make decisions. Similarly, for the protein-only IGN model, all the ligand information in the ligand graphs $G_l = (V_l, E_l)$ and the distance information in the original pocket–ligand graphs $G_{pl} = (V_{pl}, E_{pl})$ were masked by zeros.

As can be seen in Table 6, the IGN model trained on the ligand-only or protein-only dataset performs worse than that trained on the original complex dataset regardless of the best model or the average model. For the corresponding best models determined by the validation loss (MSE metric) minimum criterion, the performance of the ligand-only or protein-only model averagely dropped by 6–10% compared with its original counterpart in terms of various metrics, and it is the same for the average models from three independent runs, implying that our IGN model can utilize both protein and ligand information instead of either to make more accurate predictions. We emphasize that we are not disqualifying the IGN models trained on ligand-only or protein-only datasets.

Performing such comparative studies is just a commonly used strategy to check whether the deep models can learn proper interaction patterns, and the conclusions could be partially predicted on the relative contents of the training and test sets. Therefore, to further test whether the high performance of the IGN model trained on the protein–ligand complexes was only contributed from the intrinsic ligand or protein bias in the PDBBind dataset, we evaluated the performances of the original best IGN model trained on the protein–ligand complexes on the other two datasets. The first dataset evolves from the original test set (the core set from PDBBind V2016), but the corresponding ligand information was masked by zeros, and the second dataset was also processed by the same procedure, but the corresponding protein information was masked. One can image that the performances of the original best IGN model on these two test datasets would be satisfactory if it makes decisions based on the ligand information or protein information only. As shown in Table 7, the performances of the original best IGN model on the two

Table 7. Performance Comparison of the Four Different Versions of the Test Set Predicted by the Optimal IGN Model

test set (core set of PDBBind V2016) ^a	R_p	RMSE	MAE	R^2
complex 1	0.837	1.220	0.940	0.687
complex 2	0.550	1.945	1.580	0.204
complex 3	0.197	3.281	2.125	−1.264
complex 4	0.757	1.503	1.150	0.525

^aComplex 1: the original test set; complex 2: the original test set without ligand information; complex 3: the original test set without protein information; complex 4: the original test set but with different docking poses.

datasets were dramatically compromised, suggesting that it does not make the predictions based on the protein information or ligand information only. Here, one may question whether the high performance of the IGN model is because of simply using protein or/and ligand information alone instead of their interactions. Based on this question, another dataset was also evaluated using the original best IGN model. This dataset (Table 7, complex 4) is the same original test set, but all the ligands have different docking poses. For each data point in this dataset, we randomly picked a docking pose with a root-mean-square deviation (RMSD) larger than 2 Å relative to the corresponding experimental pose from the redocking calculations described in “Dataset Preparation” section to form this dataset. One should believe that one deep model capable of learning proper protein–ligand interaction patterns is naturally sensitive to the scoring of the same protein–ligand pair with different poses. As can be observed in Table 7, the performance on the complex 4 dataset

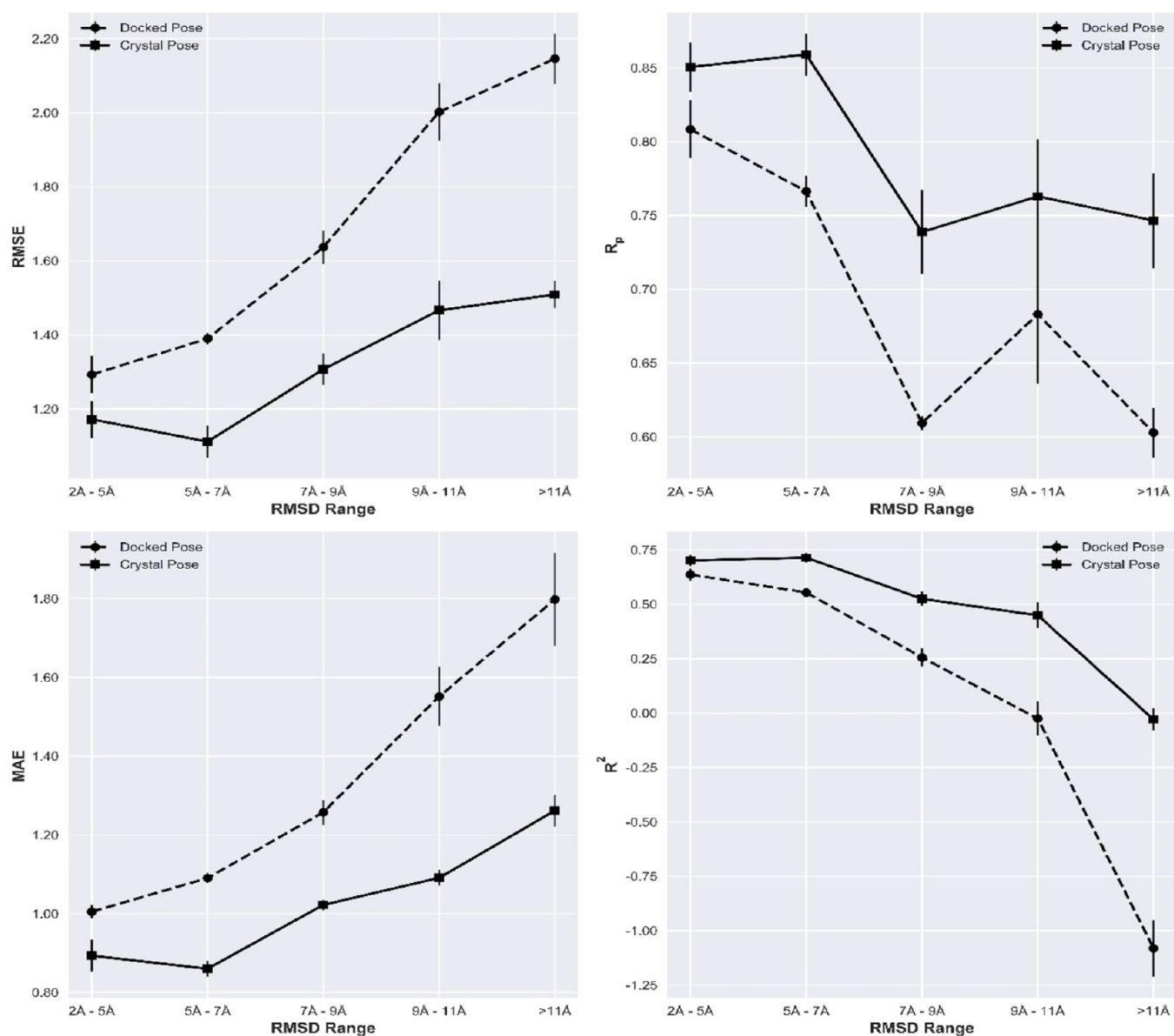


Figure 9. Performance comparison of the crystal poses and docked poses of the PDBBind 2016 core set with different RMSD bins given by the optimal IGN model.

Table 8. Performance Comparison of IGN and Glide SP on the Pose Prediction Task

run	model	training			validation			test		
		AUC_ROC	AUC_PRC	Top-1 ^a (%)	AUC_ROC	AUC_PRC	Top-1 (%)	AUC_ROC	AUC_PRC	Top-1
1	IGN	0.976	0.963	71.22	0.905	0.849	65.45	0.913	0.894	73.98
2	IGN	0.972	0.958	70.52	0.913	0.871	65.77	0.918	0.903	74.39
3	IGN	0.989	0.983	73.40	0.920	0.876	66.79	0.921	0.908	74.39
	Glide SP	0.789	0.700	67.97	0.787	0.689	67.60	0.817	0.798	78.86

^aTop-1 success rate, where the RMSD between the highest-scored conformation predicted by IGN or Glide SP and the corresponding native binding pose below 2.0 Å was regarded to be successful.

also dramatically decreased (the RMSE metric dropped by 20 % or so), indicating that the IGN model can make decisions predicted on the protein–ligand interactions. In addition, to better illustrate that the IGN model is sensitive to the different docking poses for the same protein–ligand pair, we performed more comprehensive and rigorous analyses. The docked poses in different RMSD bins were rescored using the optimal IGN model, and the predicted scores were compared with those of

the corresponding crystal poses. It should be noted that the metrics given by the corresponding crystal poses are not the metrics for the original test set shown in Table S5, and they could show some fluctuations. As can be seen in Figure 9, with the increase of the RMSD values, the performance gap between the docked poses and corresponding crystal poses would increase, especially for the three metrics including RMSE, MAE, and R^2 , demonstrating that the scoring of the

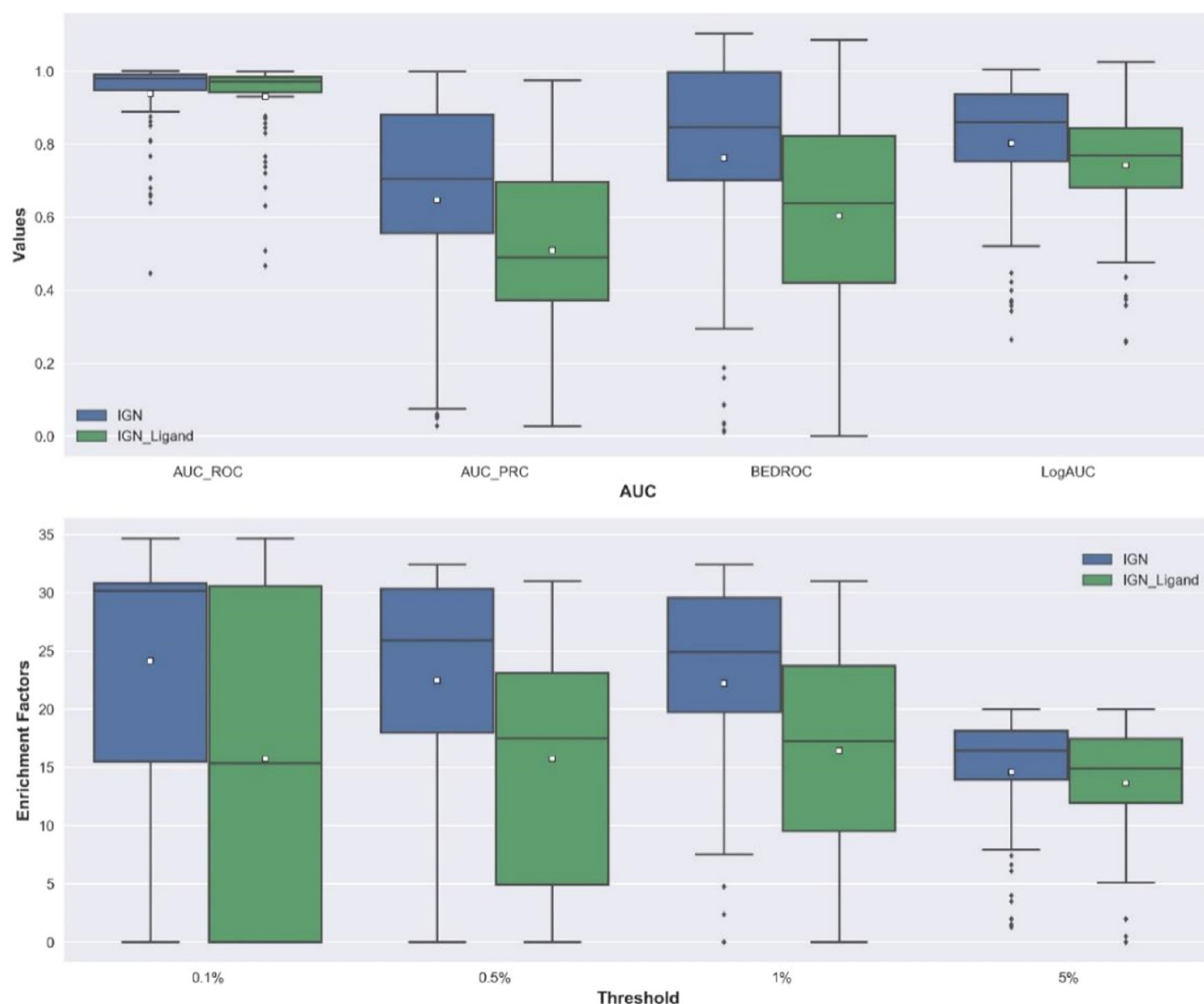


Figure 10. Performance indicators including AUC_ROC, AUC_PRC, BEDROC, LogAUC, and EFs at different thresholds including 0.1, 0.5, 1, and 5% distribution plot for each protein DEKOIS2.0 dataset given by IGN trained on the original complex-based DUD-E dataset (IGN) and trained on only the ligand-based DUD-E dataset (IGN_Ligand).

IGN model is really sensitive to the different poses for the same protein–ligand pair. Finally, we also tested the pose prediction capacity of the IGN model, as described in the “Dataset Preparation” section. As shown in Table 8, IGN yielded satisfactory AUC_ROC and AUC_PRC values above 0.900 for the test set, which is much better than those of Glide SP (both values are approximately equal to 0.800). As for the top-1 success rate, IGN is slightly worse than Glide SP (about 4 %). However, from an overall inspection, IGN is able to give reliable predictions on ligand binding poses.

Actually, the data bias, especially the ligand bias, in some large-scale VS benchmark datasets is more serious than that in the medium-sized PDBBind dataset. Therefore, similar comparative studies were also performed for the larger DUD-E and DEKOIS2.0 datasets. As shown in Figure 10, it can be recognized that the IGN model trained on the DUD-E ligand-only dataset (IGN_Ligand) performs much worse than that trained on the original DUD-E complex dataset on the external DEKOIS2.0 test set (the detailed metrics for each target in DEKOIS2.0 are available in Supporting Information

Table X5), demonstrating that the protein structure information has an essential role in the accurate predictions of IGN. Here, one interesting observation is that the AUC_ROC values for the original complex-based IGN model and IGN_Ligand model do not have apparent difference. It is well known that the AUC_ROC metric is not so appropriate for evaluating the extremely imbalanced datasets (the ratio of decoys/actives to the DEKOIS2.0 dataset is about 30/1). Therefore, such observation may be potentially caused by the insensibility and impracticability of this metric and it also reminds us that more diverse indicators should be covered when evaluating an ML-based model in addition to the simple AUC_ROC metric. Satisfactorily, it is obvious that the distributions of the AUC_PRC, BEDROC, LogAUC, and EF metrics for the 77 proteins in DEKOIS2.0 show significant difference between the original complex-based IGN model and IGN_Ligand model (Figure 10), and the posthoc Nemenyi test (significance level = 0.05) further demonstrated that the difference between the IGN model and IGN_Ligand model was statistically significant (Supporting Information Table S5).

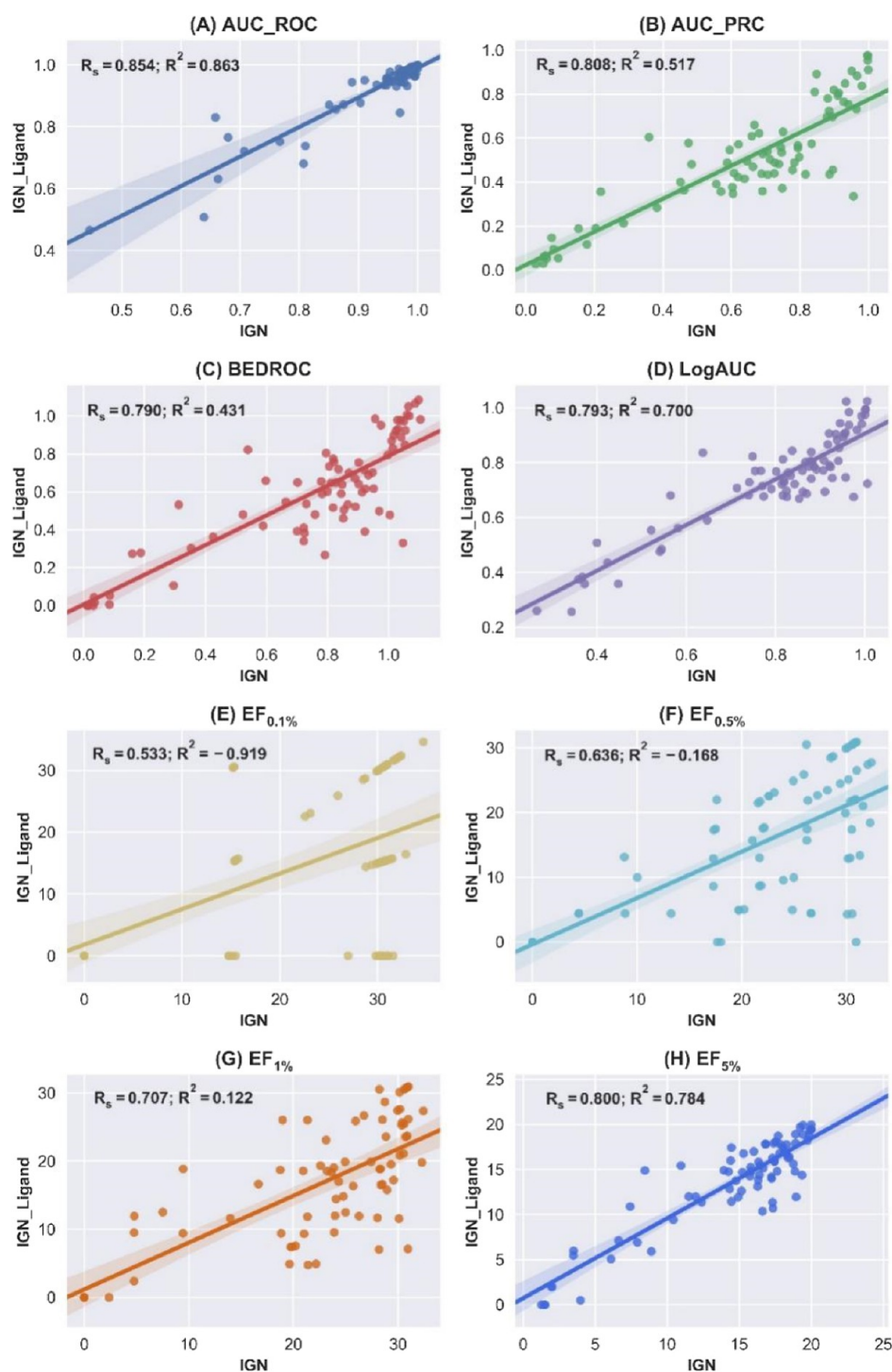


Figure 11. Correlation plot of various performance indicators including AUC_ROC (A), AUC_PRC (B), BEDROC (C), LogAUC (D), and EFs at different thresholds including 0.1 (E), 0.5 (F), 1 (G), and 5% (H) in the DEKOIS2.0 dataset between IGN trained on the original complex-based DUD-E dataset (IGN) and trained on only the ligand-based DUD-E dataset (IGN_Ligand).

Additionally, the correlations [Spearman's rank correlation coefficient (R_s) and coefficient of determination (R^2)] of various metrics achieved by the original complex-based IGN model and ligand-based IGN_Ligand model are further presented in Figure 11. It can be observed that the correlations of these metrics highly depend on the specific metrics, where the AUC_PRC, BEDROC, $EF_{0.1\%}$, $EF_{0.5\%}$, and $EF_{1\%}$ metrics show relatively low correlations, indicating that the performances of the original complex-based IGN model and

IGN_Ligand model are significantly different. However, three metrics (i.e., AUC_ROC, LogAUC, and $EF_{5\%}$) show relatively decent correlations. Here, the relatively high correlation for the AUC_ROC metric might be caused by the similar AUC_ROC values for the original complex-based IGN model and IGN_Ligand model as mentioned earlier. LogAUC is similar to the AUC_ROC, where the x-axis is plotted in the form of $\log_{10}$, and therefore, it can also give a relatively high correlation. As for the $EF_{5\%}$ metric, it is quite

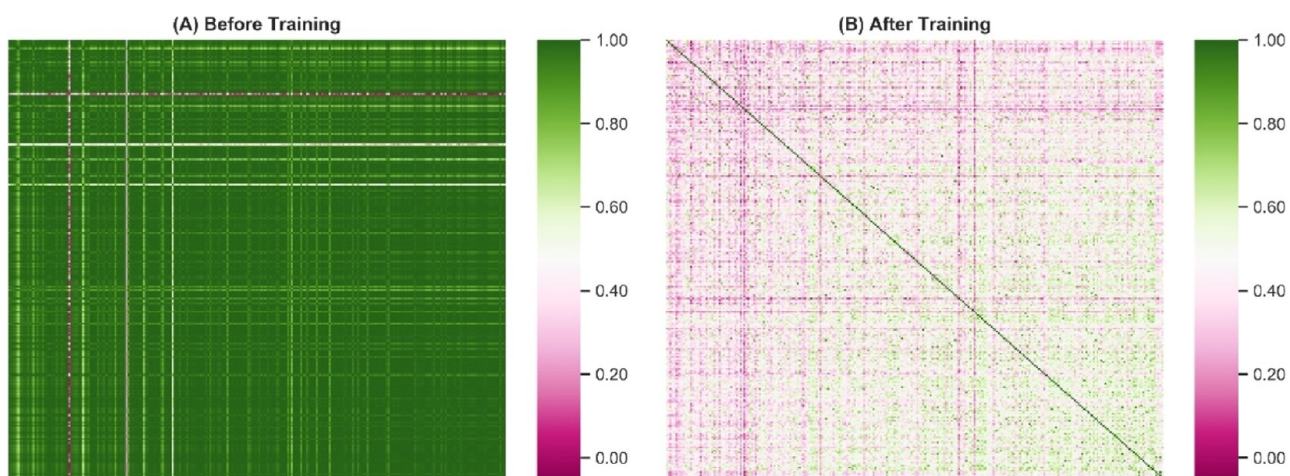


Figure 12. Similarity (R_p metric) matrix heat map plot of 262 complexes in the core set of PDBBind V2016 given by the IGN model before (A)/ after (B) training.

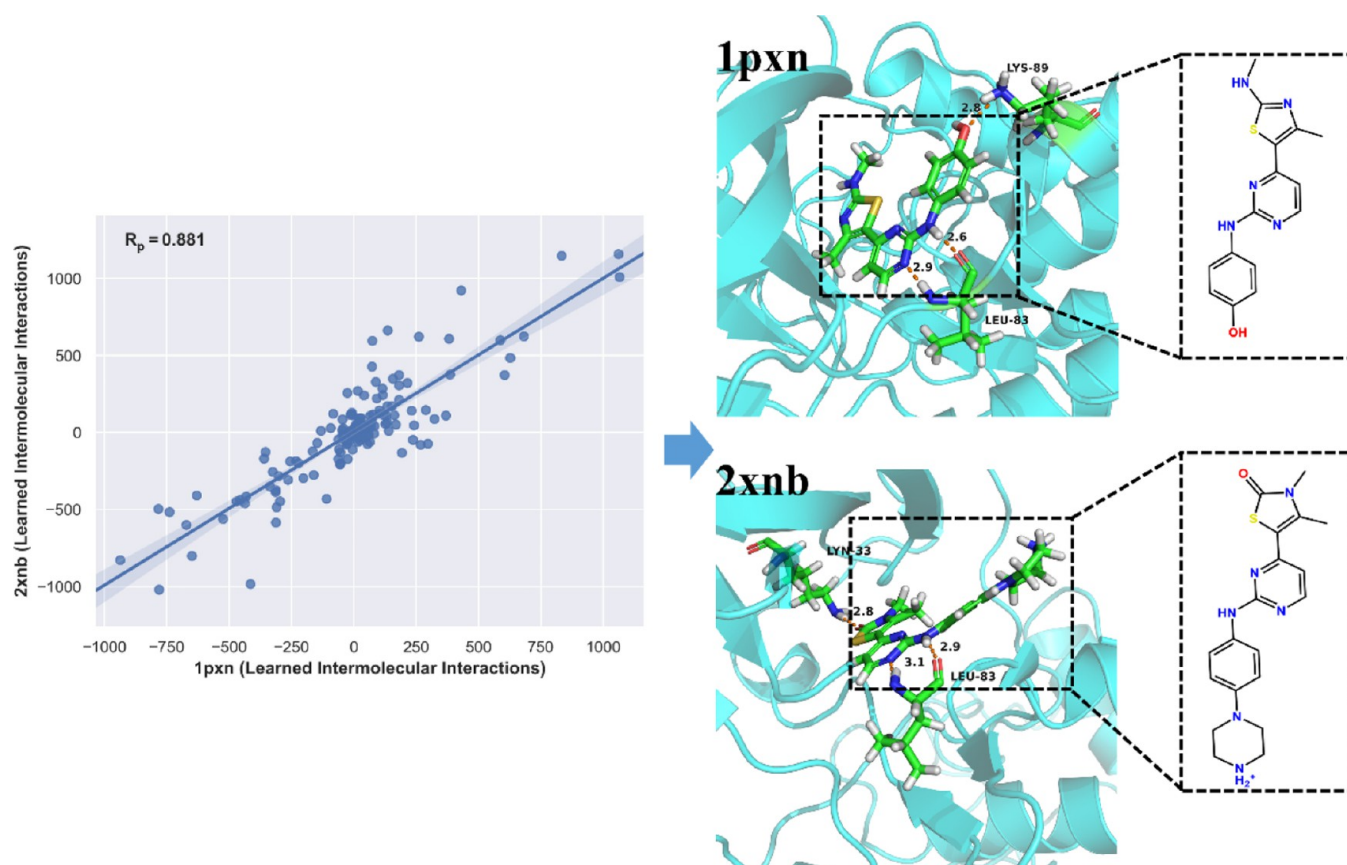


Figure 13. Correlation plot of learned intermolecular interactions between 1pxn and 2xnb complexes given by the IGN model and the ligand interaction pattern comparison of 1pxn and 2xnb complexes. Dotted line in orange represents hydrogen bond interaction.

possible that too many compounds (top 5% ranked compounds) were considered to calculate this metric, and subsequently, no obvious difference was observed for the original complex-based IGN model and IGN_Ligand model. According to the results reported by Chen et al.,^{39,40} the correlation (R^2) between the AUC_ROC values of the complex-based 3D-CNN model and those of the ligand-only 3D-CNN model for the 102 DUD-E targets is 0.99. However, the maximum correlation of the AUC_ROC values ($R^2 = 0.863$) for the IGN models is far lower than 0.99,

demonstrating that the performances of the complex-based IGN model and ligand only-based IGN model are not equivalent and the protein structure information is essential for the IGN model to make accurate decisions in structure-based VS and binding affinity predictions.

Taken together, unlike some reported 3D-CNN models that unintentionally place much lower weights on the protein atoms represented by 3D grids than those placed on the ligand atoms due to the ligand bias hidden in the training data,⁴⁰ the IGN framework was designed to automatically learn various types of

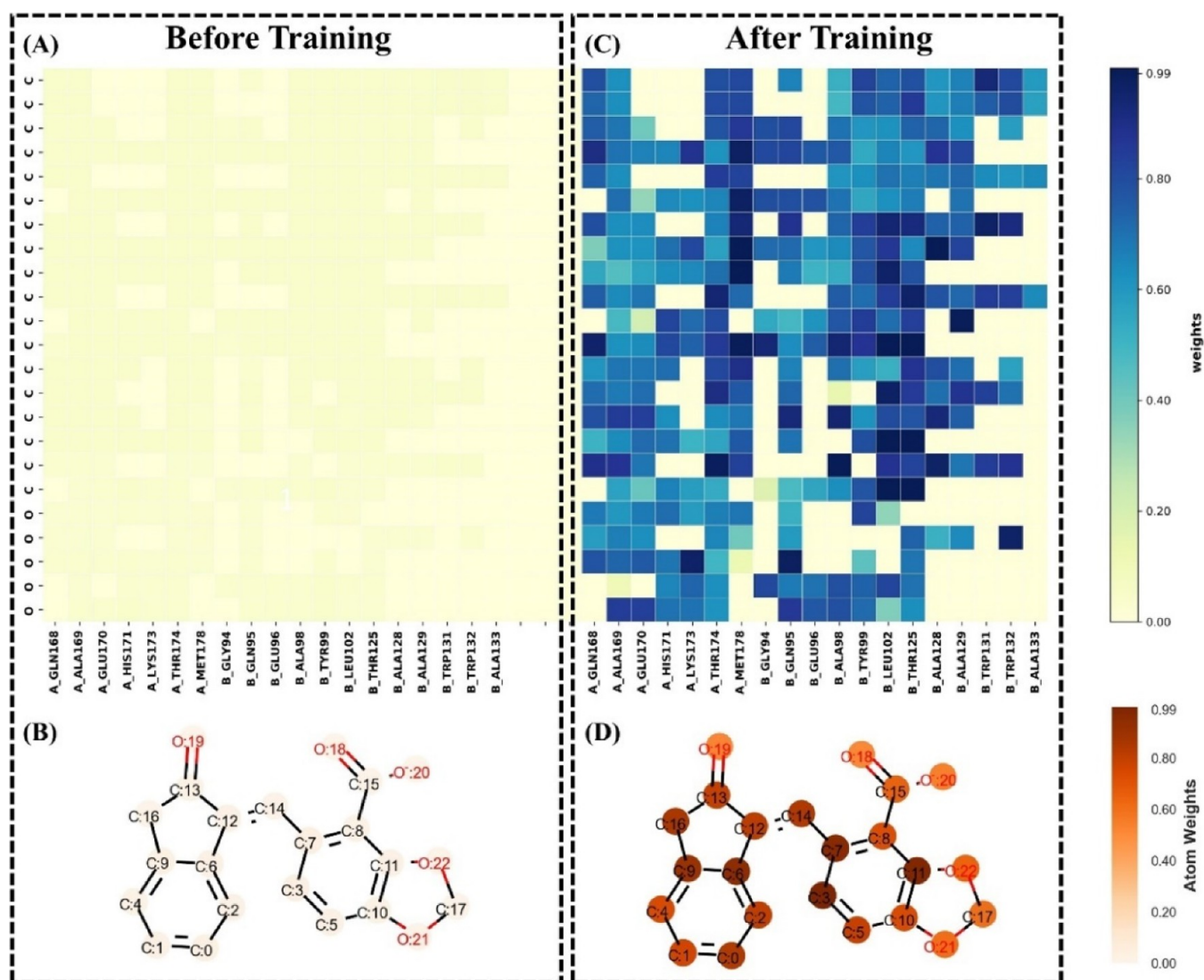


Figure 14. Interaction weight analysis for the 3zt2 complex. In (A,B), the X axis is the pocket residue information, that is, A_CLN168 represents the 186th CLN residue in chain A of the complex. The Y axis is the atom symbol information in an atom index order shown in (B,D). All the weights are maximum–minimum rescaled.

atom-pair interactions between proteins and ligands, and the subsequent graph pool aimed to extract high-level intermolecular interactions. This design philosophy was inspired by the basic physical principles for protein–ligand interactions. The lack of ligand information or protein information can lead to unobjective descriptions of the learned atom-pair interactions and the subsequent learned intermolecular interactions, and the performance of the model would be discounted accordingly.

Learned Intermolecular Interaction Visualization and Interpretation. The IGN model achieves state-of-the-art performances on various benchmarks. Therefore, it is worth exploring the interpretability of those learned intermolecular interactions raised by the “black-box” essence of DL models.⁶¹ First, the similarity analysis of the learned intermolecular interactions for the 262 complexes in the core set of PDBBind V2016 before/after training was performed. The 256-dimensional representation vector from the graph pooling layer was extracted for each investigated protein–ligand complex, and the Pearson correlation coefficient of two representation vectors was used to measure the similarity between two

complexes. As shown in Figure 12A, it can be observed that the similarity scores of the learned intermolecular interactions for most complex pairs are close to 1.0 before model training, suggesting that the interaction patterns of various complexes are almost identical and the IGN model before learning cannot well distinguish them. Nevertheless, the similarity matrix of the complexes shows different levels of chaos after model training (Figure 12B), demonstrating that different complexes may share different interaction patterns and the model can well distinguish these complexes through the learned interaction patterns, which well accords with the established domain knowledge that different ligands generally form different interactions with their targets. Here, it can be recognized that some complex pairs still share high similarity in their learned intermolecular interactions after model training. One potential explanation is that these complex pairs originally share similar interaction patterns. To further check this assumption, the binding affinity data for the complex pairs with a similarity score higher than 0.85 were extracted and are listed in Supporting Information Table S6. It can be observed that a couple of those highly correlated complex pairs share

almost the same PDB entry IDs, such as 1h23/1h22, 4ivb/4ivc, and 4de3/4de1. Generally speaking, such complex pairs were reported by the same publications and share the same protein structures and analogous coligands. Therefore, it is no surprise that the learned intermolecular interactions for them are highly correlated.

To further analyze the interaction patterns of these highly correlated complex pairs, the complex pair 1pxn/2xnb with a similarity score of 0.881 was taken as an example. The ligand interactions were identified by Schrödinger software (version 2019) and visualized by PyMOL software (version 1.8.2.2). As presented in Figure 13, the complex pair 1pxn/2xnb shares similar coligands with the *N*-phenylpyrimidin-2-amine scaffold. As for the interaction patterns, the LEU-83 residues in 1pxn/2xnb form hydrogen bonds with the same nitrogen atom of the *N*-phenylpyrimidin-2-amine scaffold. In addition, another hydrogen bond is formed between LYS-89 and the oxygen atom of the coligand in 1pxn. On the contrary, an additional hydrogen bond is formed between the neutral LYS-33 (LYN-33) and the oxygen atom of the coligand in the 2xnb complex. These analyses indicate that both 1pxn and 2xnb share similar protein–ligand interactions which are mainly dominated by the hydrogen bond between LEU-83 and the nitrogen atom in the *N*-phenylpyrimidin-2-amine scaffold. The highly correlated learned intermolecular interactions for 1pxn and 2xnb also provide an evidence for such similar interactions. A few more examples are available in Supporting Information Figure S1.

To better uncover how the ligand–protein atom–pair interactions contribute to the final prediction, the absolute weights for each ligand–protein atom pair of the 3zt2 complex calculated in the “graph pooling module” were used for analysis. Here, we visualized the atom–residue interaction weights rather than the atom–atom interaction weights in Figure 14 for a better view. For each atom–residue interaction weight, a compromised calculation way that averages the absolute weights of all possible ligand–protein atom–atom interaction in the atom–residue was assigned. Here, some weights of atom–residue pairs could be zeros because none of the atom–atom pairs was covered within 8 Å for that atom–residue pair. As shown in Figure 14A,C, the average interaction weights for each atom–residue interaction tended to be the same before IGN model training, but it showed some patterns after IGN model training, indicating that the learning of an IGN model is trying to detect some interaction patterns for a given complex. Here the average atom–residue interaction weights in Figure 14A,C imply that the residues THR174, MET178, LEU120, THR125, and TRP132 have stronger interaction weights with the ligand atoms than the others. In addition, the interaction weights for each ligand atom were visualized by averaging its interaction weights with each residue (Figure 14B,D). As can be seen, the learning of the IGN model assigned more interaction weights to the two benzene systems compared with the other components in the ligand. The ligand interaction visualization of 3zt2 in Schrödinger (version 2019) demonstrates that TRP132 and the benzene system in the 2,3-dihydro-1*H*-indene component of the ligand can form a strong π – π stacking interaction, and THR174 and the ketonic oxygen in the carboxylic acid of the ligand can form a strong H-bond interaction (data are not shown), which partially support our finding in the interaction weight analysis. All in all, these visualization results show that what the information learned by the IGN model shows good agreement with the well-established expert knowledge.

CONCLUSIONS

For many years, many ML-based predictive models were stranded on the simplified representations for the protein–ligand interactions such as hand-crafted descriptors represented by atom-pair counts, 1D sequences for proteins, or 2D graphs for ligands. It seems hard for them to learn the intricate molecular interactions in 3D space in an end-to-end manner. Here, we proposed IGN, a novel end-to-end deep graph learning framework that directly learns the intermolecular interactions from the 3D structures of protein–ligand complexes. In the IGN model, both the structural information in 3D space and chemical information were encoded into the molecular graphs to comprehensively characterize protein–ligand interaction patterns. A series of independent graph convolution modules and task layer module were well designed and stacked to gradually extract the high-level intermolecular interactions from the raw inputs. Various benchmark testings on the binding affinity prediction and structure-based VS experiments demonstrated that IGN consistently yields the best performance among various state-of-the-art methods. More importantly, the additional experiments based on the ligand-only, protein-only datasets, and the same datasets with different docking poses highlight that the IGN model can learn the key features in protein–ligand interactions but not just memorize certain biased patterns hidden in the datasets. Finally, the visualizations of the high-level learned intermolecular interactions also demonstrated the good interpretability of the IGN model.

EXPERIMENTAL SECTION

In this section, we will introduce the detailed frameworks of the IGN model which mainly includes five modules: graph representation module, intramolecular graph convolution module, intermolecular graph convolution module, graph pooling module, and task network module (Figure 1).

Graph Representation Embedded with Chemical and 3D Structural Information. As described above, a total of three molecular graphs are used to encode the intramolecular interactions and intermolecular interactions of a protein–ligand complex, namely, ligand graph ($G_l = (V_l, E_l)$), protein graph ($G_p = (V_p, E_p)$), and protein–ligand graph ($G_{pl} = (V_{pl}, E_{pl})$). The corresponding adjacency matrices A_{ij}^l , A_{ij}^p , and A_{ij}^{pl} are defined as follows:

$$A_{ij}^l = \begin{cases} 1, & \text{if } i, j \in \text{ligand atoms and } i, j \text{ are connected} \\ 0, & \text{otherwise} \end{cases} \quad (1)$$

$$A_{ij}^p = \begin{cases} 1, & \text{if } i, j \in \text{protein atoms and } i, j \text{ are connected} \\ 0, & \text{otherwise} \end{cases} \quad (2)$$

$$A_{ij}^{pl} = \begin{cases} 1, & \text{if } i \in \text{ligand atoms and } j \in \text{protein atoms and } d_{ij} < 8 \text{ \AA} \\ 1, & \text{if } j \in \text{protein atoms and } i \in \text{ligand atoms and } d_{ij} < 8 \text{ \AA} \\ 0, & \text{otherwise} \end{cases} \quad (3)$$

where $G_{pl} = (V_{pl}, E_{pl})$ is a bipartite graph, and each of its edge connects an atom in the protein and another atom in the ligand. The successful applications of the bipartite graphs in biological molecules have been reported.^{47,62} d_{ij} is the Euclidean distance between two

atoms, and such a definition assumes that certain interactions between two protein–ligand pairwise atoms exist within a threshold d_{ij} in G_{pl} . Here, d_{ij} was set to 8 Å for the binding affinity prediction task and 5 Å for the large-scale SBVS and pose prediction tasks. This threshold is empirical and optimizable, and some other thresholds such as 3 and 12 Å were also simply tested. However, with the comprehensive consideration of accuracy and computability, the choice of 8 Å is good overall. The intramolecular interactions in the ligands or proteins are represented by the covalent linkages between atoms, and the nonbonded atoms in the ligands or proteins are not considered for the purpose of computational efficiency. All complexes were preprocessed to the corresponding molecular graphs with node features and edge features. For G_l and G_p , a set of 2D node features and 2D edge features described in Supporting Information Table S7 are shared by them. Because protein–ligand interactions are critically governed by the spatial distances and directions between atoms in 3D space,⁶³ another set of 3D structural-related geometric features, including distance, angle statistics, area statistics and distance statistics, were designed as the extra-edge features for G_l and G_p .⁶⁴ These features were computed based on the spatial coordinates of atoms, and they are invariant to rotation and translation of protein–ligand complex, where the rotation and translation invariance is usually missing in some scoring functions represented by the 3D-CNN. The pseudocode and formulas of the algorithm are shown in Supporting Information Table S8. For G_{pl} , the initial node features were directly inherited from the intramolecular graph convolution module (Figure 1), and the edge features only include the intuitive Euclidean distance of the two connected atoms.

Intramolecular Graph Convolution Module. This module aims at learning the node representations for G_l and G_p , which are mainly determined by the chemical environment of the connected atoms and bonds. Inspired by the works of Xiong et al.⁶⁵ and Wang et al.,⁶⁶ the learning process of this module contains three steps: step 1: get the initial representations of nodes; step 2: update the representations of nodes through k iterations of message passing; step 3: extract the final representations of the nodes for G_l and G_p . The general symbols used in this module are listed as follows: a_i represents the initial atom features with the dimension of $F1$ for atom i ; b_{ij} represents the initial bond features with the dimension of $F2$ for bond ij , and subscript ij represents the message sent from node i to node j (i.e., from a source node to a destination node); h_i^t is the representation of node i at time step t ; $N_{(i)}$ represents the neighboring nodes of node i ; D is the dimension of the hidden state for the nodes; D' is the input size of gated recurrent unit (GRU); $\parallel$ is the concatenation operation; LeakyReLU, ELU, and ReLU are the nonlinear activations; BN is the batch normalization operation. The details for the three steps of this module are given below:

Step 1: get the initial representations of nodes.

$$a_i^{\text{new}} = \text{LeakyReLU}(w_1 a_i), \quad w_1 \in \mathbb{R}^{D \times F1} \quad (4)$$

$$b_{ij}^{\text{new}} = \text{LeakyReLU}(w_2 [a_i \parallel b_{ij}]), \quad w_2 \in \mathbb{R}^{D \times (F1 + F2)} \quad (5)$$

$$s_{ij} = \text{LeakyReLU}(w_3 [a_i^{\text{new}} \parallel b_{ij}^{\text{new}}]), \quad w_3 \in \mathbb{R}^{1 \times 2D} \quad (6)$$

$$\alpha_{ij} = \frac{\exp(s_{ij})}{\sum_{k \in N_{(i)}} \exp(s_{ik})} \quad (7)$$

$$m_i = \text{ELU} \left(\sum_{k \in N_{(i)}} \alpha_{ik} w_4 b_{ik}^{\text{new}} \right), \quad w_4 \in \mathbb{R}^{D' \times D} \quad (8)$$

$$h_i^0 = \text{ReLU}(\text{GRU}(m_i, a_i^{\text{new}})) \quad (9)$$

where a_i^{new} and b_{ij}^{new} are generated from the linear transformation of the initial atom features a_i and bond features b_{ij} , followed by the nonlinear activation according to eqs 4 and 5. Then, the unnormalized attention score s_{ij} between two connected atoms i and j are calculated by eq 6. The softmax function in eq 7 is applied to normalize the

attention scores on each atom's connected bonds. Next, a message function defined in eq 8 is used to calculate the weighted incoming message m_i for atom i . Finally, a GRU followed by nonlinear activation is used as the atom update function to fuse the transformed initial atom features a_i^{new} and accept message m_i into the initial representation of atom i : h_i^0 .

Step 2: update the initial representations of nodes through k iterations of message passing

$$s_{ij}^l = \text{LeakyReLU}(w_1^l [h_i^{l-1} \parallel h_j^{l-1}]), \quad w_1^l \in \mathbb{R}^{1 \times 2D} \quad (10)$$

$$\alpha_{ij}^l = \frac{\exp(s_{ij}^l)}{\sum_{k \in N_{(i)}} \exp(s_{ik}^l)} \quad (11)$$

$$m_i^l = \text{ELU} \left(\sum_{k \in N_{(i)}} \alpha_{ik}^l w_2^l h_k^{l-1} \right), \quad w_2^l \in \mathbb{R}^{D' \times D} \quad (12)$$

$$h_i^l = \text{ReLU}(\text{GRU}(m_i^l, h_i^{l-1})) \quad (13)$$

$$h_i^l = \text{BN}(h_i^l) \quad (14)$$

The message passing used in this module mainly contains four phases: pair concatenation, attention score calculation, message aggregation, and atom hidden-state updating (Figure 2A). However, it can also be formulated into two phases: neighboring information aggregation phase and atom hidden-state updating phase under the typical message passing neural network (MPNN) framework.⁴⁵ In the neighboring information aggregation phase, atom i accepts the information from its neighbors in a weighted way calculated by the attention mechanism, as shown in eqs 10–12. s_{ij}^l is an unnormalized attention score determined by the hidden state of nodes i and j in the l th layer, and the corresponding α_{ij}^l is the normalized attention score calculated by the softmax function in the l th layer; m_i^l is the incoming message for atom i in the l th layer. In the atom hidden-state updating phase, a GRU is used to fuse the current incoming message and previously hidden state due to its larger expressive capability (eq 13), followed by a batch normalization operation with the aim of accelerating model training to obtain the new hidden state in the l th layer (eq 14).

Step 3: get the final node representations

$$h_i^{t+1} = \sum_{t=1}^k h_i^t \quad (15)$$

As we all know, the principles of the GCN fall into neighboring information aggregation, and one of the concerns is the oversmooth issue, where the representations of nodes tend to be similar with the increase of the aggregation radius (aggregation radius can be simply regarded as the number of hidden layers in the GCN). Such an issue limits the diversity of nodes and is poisonous to node representation learning. Therefore, different from previous studies which generally utilized the node representations in the last hidden layer for the subsequent task, here we adopted a mixed node representation aggregation by skipping connection to aggregate the node representations in each hidden layer into the final node representations, as described in eq 15. Such a mixed aggregation way can alleviate the oversmooth issue with the increase of the aggregation radius while improving the diversity of the nodes.

Intermolecular Graph Convolution Module. Because the intermolecular interactions in complexes are mainly determined by noncovalent/nonbonded interactions between protein atoms and ligand atoms, this module was designed to learn the edge representations that effectively describe the pairwise atomic interactions between protein atoms and ligand atoms in G_{pl} (Figure 2B). Similarly, the general symbols used in this module are listed as follows: B_{ij} represents the initial edge features of edge ij in G_{pl} ; H_i represents the initial atom features of node i in G_{pl} .

$$H_i = h_i^{t+1} \quad (16)$$

$$B_{ij}^{\text{new}} = \text{MLP}([B_{ij} || (H_i + H_j)]) \quad (17)$$

$$B_{ij}^{\text{new}} = \text{BN}(B_{ij}^{\text{new}}) \quad (18)$$

where the initial feature of node i in G_{pl} is directly inherited from the corresponding final node representation in G_l and G_p that encodes the covalent chemical environment in 3D space according to eq 16. The edge representation B_{ij}^{new} in G_{pl} is calculated according to eq 17, where the concatenation of the sum of the initial atom features of two connected atoms and the initial edge feature is fed into a multilayer perceptron (MLP) to produce the representation, and such design logic is intended to force the edges ij and ji to share the same representation. In the same way, a batch normalization operation is used to accelerate the training of deep networks (eq 18).

Graph Pooling Module. Graph pooling is aimed at extracting a global expression for molecular graphs. Driven by the basic physical principles for protein–ligand recognition, the graph pooling module is only applied to B^{new} , which can be regarded as the learned pairwise atomic nonbonded interactions to generate the intermolecular interactions for complexes (Figure 2C). In classical molecular force fields, the map from the pairwise atomic interactions into the total nonbonded interactions is a summation. However, in this study, this map is implemented by the neural network where the concrete functional form is automatically learned by the model. The general symbols used in this module are listed as follows: $\tanh$ is the nonlinear activation; E_{pl} is the set of edges in G_{pl} ; D'' is the output dimension of MLP in eq 17; MAXPOOL is the elementwise maximum over all the values of edge features.

$$G_1^* = \sum \tanh(w_1^* B_{ij}^{\text{new}}) * B_{ij}^{\text{new}}, \quad \forall ij \in E_{\text{pl}}, \quad w_1^* \in \mathbb{R}^{1 \times D''} \quad (19)$$

$$B^{\text{new}} = B_{ij}^{\text{new}}, \quad \forall ij \in E_{\text{pl}} \quad (20)$$

$$G_2^* = \text{MAXPOOL}(B^{\text{new}}) \quad (21)$$

$$G^* = [G_1^* || G_2^*] \quad (22)$$

where the output of graph pooling G^* is combined from the weighted sum pooling G_1^* and the max pooling G_2^* . Such consideration can utilize the information from multiview where the max pooling can highlight the edge features with the highest information intensity and the weighted sum pooling can exploit the total information intensity of edge features.

Task Layer Module. Following the common practices, the FCNN augmented with the batch normalization layer is employed as the task network for the final protein–ligand interaction prediction due to its ability to be a universal approximator for any nonlinear functions. The training objective is to minimize the loss function. Given a dataset with N samples, the predicted and true values of a certain sample are $\hat{y}_i$ and y_i , respectively. The loss function (MSE loss) for the binding affinity prediction task (regression task) is defined by eq 23 and that (focal loss) for the SBVS and pose prediction task (classification task) is defined by eq 24

$$\text{MSE loss} = \frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2 \quad (23)$$

$$\text{focal loss} = -\frac{1}{N} \sum_{i=1}^N \alpha (1 - \text{sigmoid}(\hat{y}_i))^{\gamma} \log(\text{sigmoid}(\hat{y}_i)) \quad (24)$$

Focal loss is a modified function based on a cross-entropy function to deal with the class imbalance problem, and it can downweigh the loss contributions of easily classified samples to force the model to focus on the hard classified samples in the training. In eq 24, α is the weighting factor in balancing out the importance of actives and decoys and it is set as no. of decoys/no. of all complexes; γ is the focusing

parameter to adjust the rate at which the easily classified samples are downweighted, and it was set to 2.0 in our experiment.

Dataset Preparation. A total of four datasets were used in this study, including the PDBBind V2016 (version 2016),⁶⁷ DUD-E,⁶⁸ DEKOIS2.0,⁶⁹ and LIT-PCBA.⁵⁹ For the binding affinity prediction, both the general and core sets in the PDBBind V2016 were used due to the data-hungry essence of the DL models. The recent study reported by Shen et al. demonstrated that a variety of DL models with good predictive capability in binding affinity prediction actually give worse performance in large-scale SBVS.⁵⁶ Therefore, the performance of our IGN model on SBVS was tested using the large-scale DUD-E, DEKOIS2.0, and LIT-PCBA datasets. More concretely, DUD-E and DEKOIS2.0 are multiple-target datasets and they were used for constructing the target-free VS model. LIT-PCBA is a more challenging dataset, where its ratio of decoys/actives is higher than 1000 so that it was used for mimicking the real VS scenarios. LIT-PCBA is a target-specific dataset, and we constructed the target-specific VS model accordingly. More details about these datasets can be found in the corresponding publications.^{59,67–69} Additionally, it has been reported that some ML-based SF_s models with satisfactory VS performance may have less prediction capacity in distinguishing near-native binding poses from decoy poses, implying that they are insensitive to rank the docking poses for a given protein–ligand complex.^{55,70} However, one model equipped with the ability to learn physical protein–ligand interactions should have good capability to recognize near-native binding poses. To this end, a deeper pose prediction task was performed for our IGN model using the PDBBind V2016 dataset.

For the SBVS task, the 3D binding structures of the protein–ligand complexes in the DUD-E and DEKOIS2.0 datasets generated by docking calculations were directly adopted from the previous study.⁵⁶ The actives and decoys were docked into the binding sites of the corresponding proteins using the Glide SP module of Schrödinger (version 2019), and only the top-ranked conformation was retained for each complex. In addition, the preparation of the LIT-PCBA dataset in this study also conformed to the same procedure. The binding structures in the PDBBind V2016 dataset were directly provided by the original authors and the corresponding protein structures were optimized according to the procedure reported by Shen et al.,⁵⁶ as to the pose prediction task, the redocking calculations were utilized to retain a maximum of 20 binding poses for each complex in the PDBBind V2016 dataset, and the same Glide SP module was used here. A generated pose was considered as a positive sample (near-native pose) when its RMSD from the corresponding experimentally binding structure was below 2 Å or considered as a negative sample (decoy pose) when its RMSD was above 4 Å. The samples with RMSDs between 2 and 4 Å were ignored. After the 3D binding structures were obtained, the pocket residues with the distance to the ligand in each complex smaller than 5 Å were extracted using the chimera software (version 1.10.1). Next, the pocket residues and the ligand molecule were encoded as the corresponding G_p , G_l , and G_{pl} molecular graphs as described in the Graph Representation Embedded with Chemical and 3D Structural Information section using the open-source RDKit (version: 2019.09.1) and DGL-CUDA90 (version: 0.4.3.post2) Python packages. During the preparation of molecular graphs, some complexes that could not be successfully processed by any intermediate procedure were removed from the datasets.

Model Implementation, Training, and Evaluation Details.

The IGN model was implemented by the open-source DGL-CUDA90 (version: 0.4.3.post2)⁷¹ with PyTorch (version: 1.3.1+cu92) as the backend and RDKit (version: 2019.09.1) Python packages. For the binding-affinity prediction task (regression task), the whole core set from PDBBind V2016 was taken as an external test set, 20% of the general set was randomly selected as the validation set, and the remaining was used as the training set. To further test the robustness of the IGN model, another dataset (PDBBind V2013 core set) was also used as the external test set. Because some complexes in the PDBBind V2013 core set were also contained in the general set of PDBBind V2016, these complexes were directly removed from the

original PDBBind V2013 core set, and as a result, this external test set only contained 95 complexes. For the SBVS task (classification task), two kinds of models (target-free and target-specific VS models) were trained, as mentioned earlier. In the training of the target-free VS model, the whole DUD-E dataset was randomly divided into the training and validation sets at a ratio of 4:1 in a stratified sampling manner. Given that the existence of potential hidden bias has been reported for the DUD-E dataset,^{39,40} we did not use any testing complex from the DUD-E dataset because the statistics may show varying degrees of overoptimism. On the contrary, the whole DEKOIS2.0 dataset with different sources and targets was taken as the independent test set. In the training of the target-specific VS models, three representative targets (FEN1: flap endonuclease 1, KAT2A: lysine acetyltransferase 2A, and PKM2: pyruvate kinase muscle isoform 2) whose ratios of decoys to actives were all higher than 1000 were extracted from the LIT-PCBA dataset to better mimic the real VS scenarios. For each target, the corresponding test set was directly adopted from the recommendation of the original paper and the remaining datasets were divided into the corresponding training and validation sets at a ratio of 4:1 in a stratified sampling manner. For the pose prediction task, the binding poses from the training, validation, and test sets for the binding affinity prediction task were used as the training, validation, and test poses. In order to further verify the generalization ability of IGN, the IGN models trained on the DUD-E dataset were also tested on the three challenging targets in the LIT-PCBA dataset. The detailed information of the training, validation, and test sets for each experiment is shown in [Supporting Information Table S9](#). It should be noted that the number of the complexes reported in this study shows some kind of discrepancies with the original reports for each experiment, which are the results from the unsuccessful processing of the intermediate procedures such as protein preparations, docking calculation, and duplicate removal.

Here, all the models were trained using the Adam optimizer in an early stopping way if no improvement of the validation performance was observed in successive 70 epochs (15 epochs for the SBVS and pose prediction tasks) to avoid overfitting. Since one of the main focuses of this study is the application of the IGN model, the hyperparameter setting was not explored in a wide range for each experiment due to the high cost of hyperparameter evaluation. In contrast, we selected an empirical hyperparameter setting around a small range of the hyperparameter space for each task ([Supporting Information Table S10](#)). We repeated each experiment three times with different random seeds and reported the results according to the validation set error minimum criterion. As a reference, we also provided the results from three independent runs.

As for the evaluation metrics, two main indicators including RMSE and the Pearson correlation coefficient (R_p) were used to evaluate the quality of the regression model (binding affinity prediction task). In a more diverse evaluation, the coefficient of determination (R^2) and mean absolute error (MAE) were also provided. For the SBVS task, five widely used indicators were reported, including the AUC_ROC, AUC_PRC, LogAUC, BEDROC ($\alpha = 80.5$ in this study), and EFs at different thresholds (0.1, 0.5, 1, and 5%). For the pose prediction task, three main indicators, AUC_ROC, AUC_PRC, and top-1 success rate, were considered. The descriptions of these indicators can be found in many previous publications.⁵⁶ The dataset and source code implementation of IGN are available at <https://github.com/zjujdz/InteractionGraphNet/tree/master>.

■ ASSOCIATED CONTENT

SI Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.jmedchem.1c01830>.

Average metrics from three independent runs with different random seeds for the IGN model; performance comparison of the IGN model with (IGN) and without (IGN_2D) 3D atomic coordinate-related geometric features; p -values of different metrics on DEKOIS2.0

determined by the Friedman test for three compared models; p -values of different metrics on DEKOIS2.0 determined by posthoc Nemenyi test for three different model pairs; p -values of different metrics on DEKOIS2.0 determined by the posthoc Nemenyi test for the IGN model trained on original DUD-E datasets and on the DUD-E ligand-only datasets; complex pairs with similarity scores are larger than 0.850 and corresponding binding affinity data ($pK_i/K_d/IC_{50}$); node features and edge features employed in the protein and ligand graph construction; algorithm pseudocode and formulas for generating 3D edge features based on spatial coordinates of atoms for protein and ligand graphs; detailed information of training, validation, and test sets for each experiment; hyperparameter setting for regression and classification task; and correlation plot of learned intermolecular interactions for 1bcu and 4cr complexes given by the IGN model ([PDF](#))

Various performance indicators for each target in the DEKOIS2.0 dataset given by the best RFscore-VS determined by the validation set error minimum criterion; various performance indicators for each target in the DEKOIS2.0 dataset given by the Glide SP docking program; various performance indicators for each target in the DEKOIS2.0 dataset given by IGN; various performance indicators for the test sets of FEN1, KAT2A, and PKM2 from the LIT-PCBA dataset given by the IGN model; and various performance indicators for each target in the DEKOIS2.0 dataset given by the IGN model trained on the ligand-only DUD-E dataset ([XLSX](#))

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Notes

The authors declare no competing financial interest.

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ABBREVIATIONS

2D, two-dimensional; 3D, three-dimensional; AUC_PRC, area under the precision recall curve; AUC_ROC, area under the receiver operating characteristic curve; AVE, asymmetric validation embedding; BEDROC, Boltzmann-enhanced discrimination of receiver operating characteristic curve; BN, batch normalization; CADD, computer-aided drug design; CNN, convolutional neural network; DEKOIS2.0, demanding evaluation kits for objective in silico screening; DL, deep learning; DUD-E, Directory of Useful Decoys Enhanced; EFs, enrichment factors; FCNN, fully connected neural network; FEN1, flap endonuclease; FEP, free energy perturbation; GBT, gradient boosting tree; GCN, graph convolutional network; GNN, graph neural network; GRU, gated recurrent unit; IGN, InteractionGraphNet; KAT2A, lysine acetyltransferase 2A; LogAUC, area under the semilog ROC curve; MAE, mean absolute error; ML, machine learning; MPNN, message-passing neural network; MSE, mean squared error; NN, neural network; ODDT, Open Drug Discovery Toolkit; PKM2, pyruvate kinase muscle isoform 2; QM/MM, quantum mechanics/molecular mechanics; QSAR, quantitative structure–activity relationship; R^2 , coefficient of determination; RF, random forest; RMSE, root-mean-squared error; ROC, receiver operating characteristic curve; R_p , Pearson correlation coefficient; Rs, Spearman's rank correlation coefficient; SBVS, structure-based virtual screening; SD, standard deviation; SFs, scoring functions; SMILES, simplified molecular-input line-entry system; VS, virtual screening

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